30 September 1975
SUPERSEDING
MIL-C-55607(EL)
27 August 1968

MILITARY SPECIFICATION
CONVERTER, TELEPHONE SIGNAL CV-1548( )/G

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a transistorized 12 channel signal converter designated Converter, Telephone Signal CV-1548( )/G that changes conventional low ringing frequencies and dc supervisory signals to voice frequencies and converts conventional 2-wire telephone systems into 4-wire telephone systems for use with Multiplexer equipment. (See 6.1 and 6.6)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way Partial
QQ-S-781	Strapping, Steel, Flat and Seats
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5805

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MILITARY

MIL-P-116	Preservation Packing, Methods of
MIL-S-10446	Switchboard, Telephone, Manual SB-86( )/P
MIL-P-11268	Parts, Materials, and Processes Used in Electronics Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-I-46058	Insulating Compound, Electrical (for Coating Printed Circuit Assemblies)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-189	Rack, Electrical Equipment 19 Inch and Associated Panels
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipmant
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MS 35276	Screws
MS 35338	Washer, Lock

DRAWINGS

ELECTRONICS COMMAND

SC-GL-323244	Gage for Converter, Telephone Signal CV-1548/G
SC-DL-528500	Converter, Telephone Signal CV-1548( )/G

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SM-B-528523	Converter, Telephone Signal, CV-1548/G Test Requirement
SM-B-528558	Panel Assy 18A2, Generator, 20 & 1600 Cycles, Test Requirement
SM-C-528566	Cover, Front (outer)
SM-B-528626	Mounting Bracket
SC-DL-528640	Panel Assembly, Dummy Test
SM-B-528696	Washer
SM-B-528719	Panel Assy 18A3B, Channel Unit, with One Way Plug Supervision, Test Requirement
SM-B-528731	Power Supply 18A1A, Converter, Test Requirement
SC-DL-531015	Cable Assy, Power, Electrical CX-11173( )/U
SC-DL-531022	Cable Assy, Special Purpose, Electrical CX-9088( )/U

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM 11-5805-367-12	Operator and Organizational Maintenance Manual

(Copies of specifications, standards, and drawings, required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification, and of Drawing and Data List SC-DL-528500, including the requirements for parts, materials, and processes thereon.

3.3 Parts, Materials and Processes; General. In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4)

3.4 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.5 Marking.

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3.5.1 General. Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be Group 1 as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers. Each Converter, Telephone Signal CV-1548( )/G shall be serial numbered.

3.6 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.20).

3.7 Printed Wiring Assemblies. Printed wiring assemblies shall be in accordance with the applicable paragraph in MIL-P-11268 and shall be conformally coated in accordance with MIL-STD-275 with a coating material conforming to MIL-I-46058, type ER (Epoxy).

3.8 Electrical Characteristics.

3.8.1 Input Power. The equipment shall meet all its electrical specifications when the power supply is energized from 115 volts ac ± 5%, 50 Hz ± 5% to 400 Hz ± 5%. Limit values are: 47.5 Hz, 109.5 volts ac; 47.5 Hz, 121 volts ac; 420 Hz, 109.5 volts ac; 420 Hz, 121 volts ac. Maximum power input shall not be more than 30 watts with no ringing, and 80 watts with 12 channels ringing. (Compliance with these requirements shall be determined during the test of 4.8.3, 4.8.4, 4.8.5, and 4.8.6).

3.8.2 20 Hz Generator.

3.8.2.1 Frequency Stability. The frequency shall not deviate beyond the frequency range 18 Hz to 22 Hz under no load to full load under all specified environmental conditions of this specification (see 4.8.5). Full load means all 12 channels ringing.

3.8.2.2 Output Level. The generator shall provide not less than 75 volts rms simultaneously across four, two wire loop circuits. Each loop circuit shall be assumed to have an impedance of approximately 2500 ohms resistive. Load voltage shall not exceed 140 volts peak when the rung line is terminated with 10,000 ohms. The ring side of the two wire output shall be grounded during signalling (see 4.8.5).

3.8.2.3 Circuit Protection. Isolation between lines shall be maintained during ringing from switchboard or converter. The 20 Hz generator shall be protected from short circuits occuring on lines during ringing, including continuous ringing (see 4.8.5).

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3.8.3 1600 Hz Generator.

3.8.3.1 Frequency Stability. The frequency shall not deviate beyond the frequency range of 1570 Hz to 1630 Hz under no load to full load under all environmental conditions of this specification (see 4.8.4).

3.8.3.2 Output Level. The generator shall provide 0.360 volts ac ±10% to all twelve lines simultaneously. A control shall be provided with the capability of adjusting the output level 7dB above and 7dB below the nominal value. The output of each channel shall be -15dBm ± 1dB into a 600 ohm load (see 4.8.4).

3.8.4 20 Hz Detector.

3.8.4.1 20 Hz Sensitivity. The receiving sensitivity of the equipment to ringing signals shall be such that satisfactory operation is obtained with 16 Vrms 15-25 Hz sine wave input (see 4.8.2).

3.8.4.2 Circuit Protection. The detector shall withstand the output of a 20 Hz generator having an open circuit voltage of 140 volts ac peak and an internal impedance of 500 ohms (see 4.8.1).

3.8.5 1600 Hz Receiver.

3.8.5.1 Receiving Sensitivity. The receiving sensitivity shall be sufficient for satisfactory operation of the converter when a 1600 ± 30 Hz at -25dBm ac signal is received (see 4.8.3).

3.8.5.2 Frequency Selectivity. The receiver shall not respond to signals below 1450 Hz or above 1750 Hz at +5dBm (see 4.8.3).

3.8.5.3 Guard Circuit. Time delay guard circuits associated with the receiver circuitry to prevent spurious signalling by voice frequency signals shall have a minimum delay of 175 milliseconds and not more than 800 milliseconds (see 4.8.6).

3.8.6 Longitudinal Balance. The longitudinal balance of each terminal pair shall exceed 40dB over the frequency range from 300 Hz to 3500 Hz (see 4.8.8).

3.8.7 Hybrid Loss Characteristics.

3.8.7.1 Transmission. The transmission loss for all channels shall be 4.25 ± 0.25dB at the frequency of 1000 Hz when the hybrid is properly terminated (see 4.8.7.1).

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3.8.7.2 Transhybrid. The transhybrid loss for all channels from 4 wire receive to 4 wire transmit, shall exceed 30dB over the frequency range of 300 to 3500 Hz when the two wire output is terminated into 600 ohms 1% load, and shall exceed 7.5dB over the frequency range 300 to 3500 Hz when the two wire output pair is unterminated. (See 4.8.7.2).

3.8.8 Channel Frequency Response.

3.8.8.1 Four Wire-To-Two-Wire Mode. The insertion loss for each transmission path shall not exceed 4.5dB in the frequency range 300 to 3500 Hz. Signal level on the two wire pair shall be 0 dBm, and on the four wire receive pair +1 dBm. (See 4.8.9.1)

3.8.8.2 Four Wire-to-Four-Wire Mode (Patch-Thru). The insertion loss or gain for each transmission path shall not exceed 0.25 dB in the frequency range 300 to 3500 Hz. Signal levels shall be -4 dBm. (See 4.8.9.2)

3.8.9 Signal to Crosstalk Plus Noise.

3.8.9.1 1000 Hz Signal. When a 1000 Hz, 0 dBm tone is on the two wire terminals on any channel, the four-wire outputs of all unmodulated channels shall not exceed -60dBm (See 4.8.10.1).

3.8.9.2 20 Hz Signal. When a 20 Hz, 90 volt square wave signal is on the two wire terminals of any channel, the four wire outputs measured with FIA weighting of all unmodulated channels shall not exceed 24 dBa (See 4.8.10.2).

3.8.9.3 1600 Hz Signal. When a 1600 Hz, +5dBm tone is on the four wire receive terminals of any channel, the two wire outputs of all unmodulated channels shall not exceed -60dBm (See 4.8.10.3).

3.8.10 Impedance. The impedance of each terminal pair shall be 600 ohms $\pm$ 10% with a phase angle no greater than $\pm$ 10° over the frequency range 300 Hz to 3500 Hz. (See 4.8.12)

3.8.11 Harmonic Distortion. Any harmonic products which may be produced by transmission through the converter shall be at least 35dB below any fundamental within the frequency range of 300 Hz to 3500 Hz being transmitted at +1 dBm. (See 4.8.11)

3.8.12 One-Way Plug Supervision.

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3.8.12.1 Operating Requirements. The one-way plug supervision mode of operation in Converter, Telephone Signal CV-1548( )/G, as described in terms of actions of switchboard operators at Station A and Station B using Switchboard SB-86/P circuits (MIL-S-10446) and SB-86/P current limit requirements applied to Station A and B, shall be as described in 4.8.13a, b, c, d, and e, and meet the requirements listed below:

a. Idle condition (Tone on). No plugs inserted at either switchboard. All signals are BLACK.

b. Seizure (Tone off). Station A inserts call plug into the civilian trunk jack, and trunk signal at B operates to WHITE.

c. Connect and reseizure. Insertion of a call plug by Station A into the civilian trunk circuit shall operate the proper trunk signal at Station B. After this call is answered the cord signal at Station B shall operate to WHITE when the calling cord at Station A is removed, and reinserted. The trunk signal at Station B shall operate to WHITE after the answer cord is pulled down.

3.8.12.2 Originate Current. With mode switch in "OR" position, the channel unit shall operate when a 1000 ohm, 1 watt resistor is connected across tip and ring of the line. The channel unit shall not operate when a 10,000 ohm ½ watt resistor is connected between tip and ring (see 4.8.13f). **"Operate" means 1600Hz on the line.**

3.8.12.3 Idle Condition Two-Wire Termination. For the "OR" mode, the CV-1548( )/G will automatically provide the maximum transhybrid loss when no call is in progress (see 4.8.7.2).

3.8.12.4 Circuit Protection. The plug supervision circuits shall withstand, without damage, the output of a 20 Hz generator having an open circuit voltage of 140 volts ac peak and an internal impedance of 500 ohms (see 4.8.1).

3.9 Interface Compatibility.

3.9.1 Switchboard Side. Signalling circuits in the CV-1548( )/G shall be compatible with those of Switchboard Telephone, Manual SB-86/P and Telephone Set TA-312( )/PT or their equivalents. The CV-1548( )/G shall interface and operate in systems using these equipments (see 4.8.14).

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3.10 Preconditioning.

3.10.1 Electrical Preconditioning (see 4.6.1). Prior to submission to the specified quality conformance inspections, each assembled Converter CV-1548/G consisting of the chassis, plug-in panel assemblies and power supply (including all panel assemblies) shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free.

3.10.2 Bounce Preconditioning (see 4.6.2). The equipment shall be capable of meeting the requirements herein, without subsequent processing after subjection to the bounce preconditioning of 4.6.2. Each unit fabricated shall be subjected to this process.

3.11 Service Conditions. The equipment shall meet the following service conditions:

3.11.1 Temperature.

3.11.1.1 Low. During testing as specified in 4.11.1, the equipment shall meet the performance specified in 4.7, 4.8.2, 4.8.3, 4.8.5, and 4.8.13. After testing, the equipment shall meet full specification performance for these measurements.

3.11.1.2 High. During testing as specified in 4.11.2, the equipment shall meet the performance specified in 4.7, 4.8.2, 4.8.3, 4.8.5, and 4.8.13. After testing, the equipment shall meet full specification performance for these measurements.

3.11.2 Humidity. The equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials, and shall meet the performance specified in 4.7, 4.8.2, 4.8.5, 4.8.6, and 4.8.10 during the operating times specified in 4.12.

3.11.3 Altitude (elevation). The equipment shall meet full specification performance for the measurements of 4.7 both during and after testing as specified in 4.13.

3.11.4 Rain. After testing as specified in 4.14, the equipment shall meet the performance requirements of 3.14 and the interior shall not contain any evidence of water penetration.

3.11.5 Vibration, Bounce and Shock.

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3.11.5.1 Vibration (Internal). The equipment shall meet the performance specified in 4.7 when tested in accordance with 4.15. The equipment shall exhibit no visible damage after the test.

3.11.5.2 Shock, Bounce, Vehicular. The equipment shall meet the performance specified in 4.7 when tested in accordance with 4.16. At the completion of the test, the equipment shall not be damaged except for paint scrapes.

3.11.5.3 Shock (Bench Handling). The equipment shall meet the performance specified in 4.7 when tested in accordance with 4.17. No physical damage shall be evident after the test.

3.11.6 Salt fog. After the salt fog test of 4.19, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation (see 6.3) of metal. Electrical operation is not required.

3.11.7 Fungus. After the fungus test of 4.18, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3) as evidenced by growth colonization (see 6.3) (which includes branching (see 6.3) and sporulation (see 6.3)) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3) shall not be included in this evaluation.

3.12 Electromagnetic Compatibility (see 4.9). The equipment shall meet the electromagnetic interference test requirements of Class CE, Table A-I-MIL-E-55301, Notice 4 (EL), MIL-STD-461A listed below as modified herein:

CE04
CS02
CS06
RE02 (Limits as modified by Figure 19 for broadband emissions)
RE02.1
RS03.2

RE02, RE02.1, and RS03.2 shall be conducted from 14 KHz to 1 GHz.

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3.13 Subassembly Electrical Test. The CV-1548( )/G and panel assemblies including spares listed below shall meet the test requirements of their respective test requirement specification drawing as specified in 4.10:

Converter, CV-1548( )/G	(TRS-Test Requirements)	SM-B-528523
18A1A	Power Supply	SM-B-528731
18A2	Panel Assembly Generator, 20 and 1600 Hz	SM-B-528558
18A3B	Panel Assembly, Channel Unit, with Plug Supervision	SM-B-528719

3.14 Operation Inspection. Each channel shall convert a 20 Hz or dc signal from ac switchboard or telephone to a 1600 Hz signal for transmission over the voice frequency channels of multiplex equipments. The 1600 Hz ac signal from multiplex equipments shall be converted to a 20 Hz or dc signal in each channel before reaching a receiving telephone or switchboard. Each channel shall have the following modes of operation: (1) 20 Hz signalling, 2 wire; (2) plug supervision, 2 wire; (3) no signalling, 2 wire (Hybrid only, in use); and (4) no signalling, 4 wire (channel patched straight through). (See 4.7)

3.15 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:

Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.

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Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-Tapping screws.

4. Quality Assurance Provisions.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery; see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections:

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.23).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article units. The contractor shall furnish five first article units of the complete CV-1548( )/G converter.

4.3.2 First Article inspection. The first article inspection shall consist of the inspections specified in Table 1 and shall be performed in the order specified therein.

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TABLE I. ORDER OF FIRST ARTICLE INSPECTION

Inspections (Note A)	Reqt Para	Test Para	Number of Samples - Five				
			Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
1. Inspections covered by subsidiary documents.		4.4	X	X	X	X	X
2. Bounce Preconditioning	3.10.2	4.6.2	X	X	X	X	X
3. Electrical Preconditioning	3.10.1	4.6.1	X	X	X	X	X
4. Group A	SEE TABLE II		X	X	X	X	X
5. Group B	SEE TABLE III		X	X	X	X	X
6. EMI	3.12	4.9					1
7. Environmental							
Low Temperature	3.11.1.1	4.11.1	1				
High Temperature	3.11.1.2	4.11.2	2				
Rain	3.11.4	4.14		1			
Altitude (elevation)	3.11.3	4.13		2			
Shock (bench handling)	3.11.5.3	4.17				1	
Vibration (resonance search)	3.11.5.1	4.15				2	
Moisture Resistance	3.11.2	4.12	3				
Fungus (Note C)	3.11.7	4.18			1		
Salt Fog (Note C)	3.11.6	4.19		3			
Shock, Bounce, Vehicular	3.11.5.2	4.16				3	

NOTES:

NOTE A: The inspections 1 through 5, in the order show, shall be performed on all First Article units before subjecting these units to any other inspection requirements specified in the table.

NOTE B: The numbers in the unit columns in the table covered by Inspections numbered 6 and 7 specify the order of inspections for the indicated unit.

NOTE C: The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.3.3 First Article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, materials and processes; general	3.3
Finishes	3.4
Marking	3.5
Printed wiring assemblies	3.7
Workmanship	3.15

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspections specified in 4.4, 4.6, and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly identified in paragraph 3.13 shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the operational inspection in 4.7 shall be last.

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TABLE II. GROUP A INSPECTION

Inspection	Req Para	Test Para	AQL Major	AQL Minor
Printed Circuit Boards				
Visual-Mechanical				
Case and Covers (Rear, Top and Front)	3.15	4.21	2.5	10.0*
Sub-Assemblies				
Power Supply (18A1A)	3.15	4.21	1.0	4.0*
Tone Generator (18A2)	3.15	4.21	1.0	4.0*
Channel Unit (18A3B)	3.15	4.21	1.0	4.0*
Converter CV-1548, Final Assy	3.3 3.15	4.21	2.5	10.0*
Electrical				
Sub-Assemblies (Electrical Test)				
Power Supply (18A1A)	3.13	4.10	1%	**
Tone Generator (18A2)	3.13	4.10	1%	**
Channel Unit (18A3B)	3.13	4.10	1%	**
Converter CV-1548				
20 Hz Generator				
Frequency Stability	3.8.2.1	4.8.5	.65%	**
Power Output	3.8.2.2	4.8.5	.65	**
Circuit Protection	3.8.2.3	4.8.5	.65%	**
1600 Hz Generator				
Frequency Stability	3.8.3.1	4.8.4	.65%	**
Power Output	3.8.3.2	4.8.4	.65%	**
20 Hz Detector				
20 Hz Sensitivity	3.8.4.1	4.8.2	.65%	**
Circuit Protection	3.8.4.2	4.8.1	.65%	**
1600 Hz Receiver				
Receiving Sensitivity	3.8.5.1	4.8.3	.65%	**
Frequency Selectivity	3.8.5.2	4.8.3	.65%	**
Channel Frequency Response				
Four Wire-to-Two wire mode	3.8.8.1	4.8.9.1	.65%	**
Four Wire-to Four wire mode	3.8.8.2	4.8.9.2	.65%	**
Operational	3.14	4.7	1%	**

*Defects per Hundred Units
**All electrical defects are major

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4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. The Group B sampling plans for the AQL's listed in Table III, shall be as follows:

AQL	INSPECTION LEVEL
4.0 percent	S-4
6.5 percent	S-4

4.5.2.2 Order of Inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

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TABLE III. GROUP B INSPECTION

Inspection	Req Para	Test Para	AQL
Interchangeability	3.6	4.20	
Dimensional (subassemblies)	3.6	4.20.1	6.5%
Electrical (subassemblies)	3.6	4.20.2	6.5%
Electrical			
Harmonic Distortion	3.8.11	4.8.11	4.0%
1600 Hz Receiver Guard Circuit	3.8.5.3	4.8.6	4.0%
One Way Plug Supervision			
Operating	3.8.12.1	4.8.13a thru e	4.0%
Originate Current	3.8.12.2	4.8.13f	4.0%
Circuit Protection	3.8.12.4	4.8.1	4.0%
Idle Condition two-wire termination	3.8.12.3	4.8.7.2	4.0%
Hybrid Loss Characteristics			
Transmission	3.8.7.1	4.8.7.1	4.0%
Transhybrid	3.8.7.2	4.8.7.2	4.0%
Longitudinal Balance	3.8.6	4.8.8	6.5%
Signal to Crosstalk Plus Noise	3.8.9	4.8.10	6.5%
Impedance	3.8.10	4.8.12	6.5%
Interface Compatibility	3.9	4.8.14	6.5%

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

TABLE IV. GROUP C INSPECTION

Inspection	Req Para	Test Para
Sub-Group 1		
Service condition:		
Temperature	3.11.1	4.11
Low	3.11.1.1	4.11.1
High	3.11.1.2	4.11.2
Moisture Resistance (Humidity)	3.11.2	4.12
Sub-Group 2		
Shock (bench handling)	3.11.5.3	4.17
Vibration (resonance search)	3.11.5.1	4.15
Shock, bounce, vehicular	3.11.5.2	4.16
Rain	3.11.4	4.14
Sub-Group 3		
Elevation	3.11.3	4.13
Salt-Fog*	3.11.6	4.19
Fungus*	3.11.7	4.18
EMI	3.12	4.9

*The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Sub-Group 1. For this sub-group, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every month. Both units shall be subjected to temperature and moisture resistance (humidity) testing in that order.

4.5.3.1.2 Sub-Group 2. For this sub-group, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every two months. One unit shall be subjected to shock (bench handling) and vibration (resonance search), the other unit shall be subject to rain and shock (bounce) testing in that order.

4.5.3.1.3 Sub-Group 3. For this sub-group, three units shall be selected from the first production lot. For subsequent Group C inspection, three units shall be selected every six months. One unit shall be subjected to elevation and salt fog testing, in that order, one unit shall be subjected to fungus testing, and one unit shall be subjected to EMI.

4.5.3.2 Order of Inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group B and Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or purchase order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Disposition of Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order, provided the sample units meet the requirements of 4.5.4.

4.6 Preconditioning (see 3.10). Each assembled CV-1548( )/G shall be Bounce Preconditioned and Electrically Preconditioned (including spare panels and power supplies), prior to submission to the Government for Group A, B, and C tests.

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4.6.1 Electrical Preconditioning. The electrical preconditioning shall be conducted per test level B of MIL-STD-781 and performed as described below. Each panel, including spares, shall also meet the requirements of 3.10, Preconditioning.

a. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued without the restart of a new 48 hour preconditioning period. A plug-in panel assembly, power supply or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48 hour operation preconditioning period prior to final electrical assembly tests by the contractor.

b. The specified normal or simulated operating signals shall be applied to the equipment during the operational preconditioning and periodic checks performed to detect any failures that may occur. During this operational preconditioning, the units shall be energized for 23 hours (consecutively) and deenergized for one hour during each 24 hour period. Lapse time clocks shall record each unit's cumulative ON time and a written record shall be kept of each unit's ON and OFF time in a log book.

c. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 144 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

d. The contractor shall perform the electrical preconditioning as described in paragraphs 4.6.1.1 and 4.6.1.2 below.

4.6.1.1 Operation of Unit During Preconditioning. The unit containing all subassemblies shall be set up on a heat-rack assembly. Then the equipment shall be subjected to the Operational Test, (4.7). After meeting the requirements of the Operational Test (4.7) the Test Circuit (Figures 1 and 20) shall be established. Channel 1 will be placed into the OR mode; all other channels will operate in the AC mode. 115 V ac,

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60 Hz, power shall then be applied for a period of 48 hours divided into two 24 hour periods. The operational test shall be repeated at the end of 24 and 48 hours. During each 24 hour period, all channels are monitored for normal operation as described in 4.7. Normal Operation during Preconditioning is defined as follows:

Channel 1 (OR mode) transmits a 1600Hz signal originating in the 18A2 Signal Generator module to the MUX Receive Terminals of Channel 2. Channel 2 (AC mode), reacting to the 1600 Hz signal, transmits a 20 Hz signal originating in the 18A2 Signal Generator module to the SWBD Tip and Ring Terminals of Channel 3. Channel 3 (AC mode) reacting to the 20 Hz signal applies a 1600 Hz signal to the MUX Receive terminals of Channel 4 (AC mode). The above described process repeats itself to and including Channel 12.

4.6.1.2 Preconditioning Test Procedures.

a. The equipment shall be checked during each 24 hour preconditioning period for normal operation. Normal operation, as defined in 4.6.1.1 is established by turning the Channel 1 mode switch to the OFF position and returning it to the OR position. During the OFF position, channels must return to a neutral state and the channel indicator lights of the test circuits must go off. When the Channel 1 mode switch is returned to the OR position, all channel indicator lights of the test circuit must return to ON and remain ON until the next test, or deenergization of the total unit takes place. If a failure occurs in any of the plug-in panel assemblies, power supplies, or chassis, it may be replaced with a previously preconditioned plug-in panel assembly, power supply, or chassis.

b. The contractor shall perform tests to demonstrate normal operation of the equipment during the periodic operational tests. After completion of preconditioning, the unit shall meet the operational inspection of 4.7.

c. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning, may be repaired and electrically retested, accepted and delivered without repeat of electrical preconditioning.

d. All electrically preconditioned plug-in assemblies, power supplies, and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group 3 requirements of MIL-M-13231.

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4.6.2 Bounce Preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.7 Operational Inspection. (loop back) On the multiplexer side of the converter, connect odd channel receive pairs to the adjacent higher number (even) transmit pairs, and odd channel transmit pairs to the adjacent higher number (even) receive pairs. (See Figure 1). On the switchboard side of the converter, connect each channel pair to a junction box. Connect power cable assembly.

a. Operate 2W/4W switches to a 4-wire (4W) position. Check each channel for electrical continuity through the switchboard cable pairs. (Receive-to-transmit, transmit-to-receive).

b. Operate 2W/4W switches to 2-wire (2W) position. Turn power on. Power indicator lamp and 20 Hz lamp shall light.

c. Set even channel mode switches to "OR" and odd channel mode switches to "TE". Put a 1,000 ohm resistor across the (T)-(R) switchboard receive pair of an even channel; check corresponding odd channel for d.c. continuity across (T)-(R) switchboard receive pair. Repeat for all channel pairs.

d. Set odd channel mode switches to "OR" and even channel modes switches to "TE". Put 1,000 ohm resistor across switchboard receive pair of an odd channel; check corresponding even channel for d.c. continuity across the switchboard receive pair. Repeat for all channel pairs.

e. Set each channel mode switch to "AC"; connect a TA-312( )/PT or equal to an even channel; connect a second TA-312( )/PT or equal to the corresponding odd channel. Ring each way and talk between telephones. Repeat for all channel pairs.

f. The results shall meet the requirement of 3.14.

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4.8 Electrical Testing.

4.8.1 Two-Wire Circuit Protection.

a. Connect equipment as shown in Figure 3.

b. Adjust oscillator to 1,600 Hz, -15 dBm.

c. Operate each channel mode switch of the unit under test (unit #2) through all its positions.

d. Apply operational check of 4.7.

e. The requirements of 3.8.4.2 and 3.8.12.4 shall be met.

4.8.2 20 Hz Sensitivity.

a. Connect the circuit as shown in Figure 4.

b. Adjust oscillator to 20 Hz and 16.0 vrms.

c. Meter shall indicate ringing.

d. Record compliance.

e. Repeat b through d for 15 Hz and 25 Hz on same channel.

f. Repeat b through e for all channels.

g. The requirements of 3.8.4.1 shall be met.

4.8.3 1600 Hz Sensitivity and Frequency Selectivity.

a. Connect circuit as shown in Figure 5.

b. Adjust a-c line voltage to 121 vrms 47.5 Hz for the first set of measurements.

c. At 1450 Hz, +5 dBm record compliance if closure occurs.

d. At 1570 Hz, -25 dBm record compliance if open circuit.

e. At 1630 Hz, -25 dBm record compliance if open circuit.

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f. At 1750 Hz, +5dBm record compliance if closure occurs.

g. Adjust a-c line voltage to 109.5V, 420 Hz and repeat c through f.

h. Repeat for all channels.

i. The requirements of 3.8.1, 3.8.5.1, and 3.8.5.2 shall be met.

4.8.4 1600 Hz Generator Power Output and Frequency Stability.

a. Connect the test circuit as in Figure 6. Adjust the a-c line voltage to 115 vrms, 60 Hz.

b. Record output power of all channels.

c. Record at the direct output of the 1600 Hz generator, rms voltage and frequency.

d. Adjust a-c line voltage to 121 vrms, 47.5 Hz and repeat b and c.

e. Adjust a-c line voltage to 109.5 vrms, 420 Hz and repeat b and c and perform the test listed in f below.

f. Adjust 1600 Hz output potentiometer to maximum and minimum settings. Record output powers of one channel. Adjust the 1600 Hz output potentiometer to normal level.

g. The requirements of 3.8.1, 3.8.3.1, and 3.8.3.2 shall be met.

4.8.5 20 Hz Generator Power Output, Frequency Stability and Circuit Protection.

a. Connect the test circuit as shown in Figure 7.

b. Set oscillator to operate all four channels. Apply a short circuit to any channel, continue ringing for a minimum of three minutes, on the selected channel.

c. While ringing record the power consumption.

d. After ringing record power.

e. Adjust a-c line power to 109.5 vrms, 420 Hz. Replace short on the selected channel with 2500 ohms

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f. Record ringing voltage and frequency of the selected channel.

g. Turn off power. Disconnect oscillator from all but the selected channel; on this channel disconnect 2500 ohm load resistor and replace with 10k ohm. Turn on power to converter. Adjust to 121 vrms at 47.5 Hz.

h. Record ringing voltage and frequency on the selected channel (b above).

i. The requirements of 3.8.1, 3.8.2.1, 3.8.2.2, and 3.8.2.3 shall be met.

4.8.6 Guard Circuit Time Delay.

a. Connect the test circuit as shown in Figure 8. Adjust a-c line to 121 vrms, 47.5 Hz.

b. Operate pushbutton switch to trigger counter and transmit 1600 Hz into channel under test. Set sensitivity of both counter inputs for reliable triggering. Adjust 1600 Hz input to +2 dBm for first measurement.

c. Push button, read counter and record value.

d. Adjust 1600 Hz input to -15 dBm.

e. Repeat c above.

f. Adjust a-c line to 109.5 vrms, 420 Hz and repeat b through e above.

g. The requirements of 3.8.1 and 3.8.5.3 shall be met.

4.8.7 Hybrid Loss Tests.

4.8.7.1 Hybrid Transmission Loss.

a. Connect the test circuit as shown in Figure 9.

b. Adjust the oscillator for +6 dBm, 1000 Hz.

c. Compute the loss in each case by the difference of readings. (M2-M1-6dB)

d. Repeat b and c above for all other channels.

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e. The requirements of 3.8.7.1 shall be met.

4.8.7.2 Hybrid Trans-hybrid Loss.

a. Connect the test circuit as shown in Figure 9.

b. Adjust the oscillator for +6 dBm, 1000 Hz.

c. The loss for each channel is computed as the difference of readings (M2-M3-6 dB).

d. Repeat for 300, 3500 Hz.

e. Remove M1 and associated resistor.

f. Repeat b, c, and d above.

g. The requirements of 3.8.7.2 shall be met.

h. Set the channel mode switch to "OR" and repeat b through g above.

i. The requirements of 3.8.12.3 shall be met.

4.8.8 Longitudinal Balance Test.

a. Align equipment for normal operation with each channel mode switch in the AC position. Refer to the longitudinal Balance Test Terminations Chart below for CV-1548( )/G terminal pair terminations for each test.

b. Connect the equipment as shown in Figure 10A. Measurements shall be made at 300 Hz, 1000 Hz, and 3500 Hz, with an input of +1 dBm (+7 dBm at V1A). Record the output at V2 in dBm. Measure all channels.

c. Connect the equipment as shown in Figure 10B. Repeat the test procedure of b above, except the output of V3 shall be recorded in dBm. Measure all channels.

d. For a randomly selected channel, repeat the measurements of b and c with the channel mode switch in "OR", "TE", and "OFF" positions for the switchboard-receive pair. (Replace V2 or V3 with Transmission Measuring Set using F1A weighting if noise is a problem).

NOTE: When testing in the originate (OR) mode, a one henry, 5 milliampere, audio choke must be placed across the tip-ring (*) pair to provide a d.c. closure, or the generator (18A2) may be disconnected.

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e. Longitudinal balance in dB at each frequency is the difference of V3-V2. The requirements of 3.8.6 shall be met.

LONGITUDINAL BALANCE TEST TERMINATIONS CHART

Measure longitudinal balance of	Connect the input to	Connect the output to	Connect 600 ohms to
SWBD receive MUX receive MUX transmit	SWBD receive* MUX receive MUX transmit	MUX transmit SWBD receive* SWBD receive*	MUX receive MUX transmit MUX receive

4.8.9 Channel Frequency Response.

4.8.9.1 Four Wire-to-Two Wire Response.

a. Connect the equipment and set the channel mode switch as shown in Figure 11.

b. For any randomly selected channel set oscillator for +7 dBm input for frequencies 300, 1100, 1900, 2700, and 3500 Hz and record the output.

c. Repeat the test with channel mode switch in the AC position.

d. The requirements of 3.8.8.1 shall be met.

4.8.9.2 Four Wire-to-Four Wire Response.

a. Connect the equipment as shown in Figure 12.

b. For any randomly selected channel set oscillator for -4 dBm input for frequencies 300, 1100, 1900, 2700, and 3500 Hz and record the output.

c. The requirements of 3.8.8.2 shall be met.

4.8.10 Signal to Crosstalk Plus Noise.

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4.8.10.1 1000 Hz Signal. Connect the test circuit as shown in Figure 13. After calibrating the transmission measuring set, apply 1000 Hz, 0 dBm signal to the SWBD receive pair of channel Group 1 through 4. Measure non-modulated channels of the same connector group. Move signal consecutively until all channels of the group have been modulated. Repeat for channel groups 5 to 8 and 9 to 12. The requirements of 3.8.9.1 shall be met.

4.8.10.2 20 Hz Signal. Connect the test circuit as shown in Figure 14. In connector channel group 1 - 4 select channel 1 and press the pushbutton switch to obtain 20 Hz ringing. With FIA weighting measure the non-modulated channels of the same connector group. Move signal consecutively until each channel of the group has been modulated. Repeat for connector groups 5 to 8 and 9 to 12. Requirements of 3.8.9.2 shall be met.

4.8.10.3 1600 Hz Signal. Connect the equipment as shown in Figure 15. Apply a 1600 Hz +5 dBm signal to the MUX receive pair. Measure non-modulated channels of the same connector group. Move signal consecutively until all channels have been modulated. Repeat for channel groups 5 to 8 and 9 to 12. The requirements of 3.8.9.3 shall be met.

4.8.11 Harmonic Distortion. Connect the equipment as in Figure 16. Set oscillator for +7 dBm for +1 dBm input for frequencies 300, 1100, 1900, 2700, and 3500 Hz. The requirement of 3.8.11 shall be met.

4.8.12 Impedance. Connect the equipment as in Figure 17. Interchange, in turn, each of the terminating resistors with the impedance bridge until all have been measured. Repeat for all channels. The requirements of 3.8.10 shall be met.

4.8.13 One-Way Plug Supervision. Connect equipment as shown in Figure 18. Place a 150 ohm resistor in series with each lead of the switchboard cable and respective terminals on Switchboard SB-86( )/P. Perform the following tests for conformance to 3.8.12. Adjust the ac line voltage to 121V, 47.5 Hz for the first set of measurements, then 109.5 V, 420 Hz for the second set of measurements. Adjust switchboard voltage normal for first set of measurements and high for second set of measurements. (See 3.8.1)

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a. Operate the SB-86( )/P Civilian Truck Line Selector Switch to M and Civilian Trunk Switch to "ON". Set other line selector switches to C. Set Telephone TA-312( )/PT switches to CBS.

b. Move phone "A" off hook. Select an "answer" cord and put this cord into the calling line jack (A). Operate the cord switch to "Talk-List" and insert the plug of the corresponding "call cord" into the channel 1 CIV trunk. The line signal at channel 1 (TE) shall operate to white. A dc meter placed in series with either of the 150 ohm resistors of channel 1 (TE) shall read from 5 to 20 ma. Restore cord switch.

c. Take another "answer" cord, operate its switch to "Talk-List" and answer the call. Put the corresponding call cord into jack of phone B. Both phones A and B shall now talk. Restore cord switch to normal (vertical position). No signal shall operate (all black).

d. Remove "call" cord from civilian trunk. The "answer" cord signal of station B shall operate to WHITE position. Reinsert "call" cord of Station A in civilian truck. Pull down answer cord of B. Trunk Signal of B shall be in WHITE position.

e. The requirements of 3.8.12.1 shall be met.

f. Remove call cord and insert 680 ohms 5 watt resistor across civilian trunk tip-ring (1000 ohms resistance consisting of one each 680 ohm resistor and two each 150 ohm resistors (R3 and R4) placed across tip-ring "OR" converter panel). Connect 10,000 ohm ½ watt resistor across civilian trunk. The requirements of 3.8.12.2 shall be met.

4.8.14 Switchboards Interface Test.

a. Connect switchboard furnished for test as shown in Figure 2. Loop-back audio on the multiplex side of converter to adjacent channel as shown.

b. Set all channel mode switches to "AC" and signal over each channel of switchboard using internal 20 Hz ringer. Each associated trunk signal shall operate correctly for ac ringdown mode at remote switchboard.

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c. Set all even channel mode switches to "OR" and odd channel mode switches to "TE". Signal over each even channel by dc plug supervision (see 4.8.13). Each odd channel associated trunk and cord signal shall operate.

d. Set all odd channel mode switches to "OR" and all even channel mode switches to "TE". Signal over each odd channel by dc plug.

e. The requirements of 3.9 shall be met.

4.9 Electromagnetic Compatibility Testing. Perform all tests in paragraph 3.12 in accordance with Notice 3 (EL), MIL-STD-462. An approved EMC test plan is mandatory prior to the start of formal qualification testing.

4.9.1 Operating Conditions. The equipment shall be operated in all signalling modes for compliance with 3.12. The following shall be considered as normal signal sources:

a. Controlled intermittent 20 Hz, 100 vrms test signal.

b. A controlled intermittent 24 volt dc, closure test signal.

Arrangements may be made for simultaneous testing of all modes provided that no two channels are simultaneously signalling.

4.10 Electrical Subassembly Test (see 3.13). The Converter CV-1548( )/G and panel assemblies, including spares, shall be tested in accordance with their respective test requirement specification drawing listed in 3.13.

4.11 Temperature Tests. All pretest electrical tests shall be performed after the units are in the chamber with the chamber door open.

4.11.1 Low Temperature. The equipment shall be subjected to Test Method 502.1, Procedure 1 of MIL-STD-810. Storage temperature shall be minus 65 +0, -5 degrees F. Storage time shall be 72 hours. Operation temperature shall be minus 25 -5 +0 degrees F. At operation temperature the test shall start within five minutes after AC power is applied to the unit. Pretest and post test data required is specified in 3.11.1.1. Performance tests shall be made during this test as specified in 3.11.1.1 and Test Method 502.1.

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4.11.2 High Temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II of MIL-STD-810. The operating temperature shall be +120, +5, -0 degrees F. The internal chamber temperature in Step 4 shall be 154 degrees F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.11.1.2.

4.12 Humidity. The equipment shall be subjected to the humidity test specified in Method 507.1, Procedure II, of MIL-STD-810, with test measurements during Step 4, and during the 24 to 28 hour period and last 5 hour period of each cycle in Step 6. Inability of the equipment to meet the requirements of 3.11.2 shall constitute failure of this test.

4.13 Low Pressure (Altitude). The equipment shall be subjected to the following test:

a. The equipment shall be placed in the test chamber. Prior to testing, the equipment shall meet the requirements for the tests specified in 3.11.3 except that auxiliary test equipment need not be inclosed in the chamber. A temperature of 73 degrees ± 18 degrees F shall be maintained during the entire test.

b. Reduce the chamber internal pressure to 87.5 mm of HG (3.44 in. of Hg or 50,000 feet above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

c. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 feet above sea level) and then perform the test of 3.11.3.

d. With the equipment not operating, return the chamber to standard ambient conditions and perform the test of 3.11.3 and visually inspect the equipment.

e. During and after the test, the equipment shall meet the requirements for the test of 3.11.3 and shall not exhibit any electrical or mechanical degradation which could in any manner prevent the equipment from meeting any requirements of this specification.

4.14 Rain Test. The equipment shall be tested as follows:

a. Dry at 130 degrees F ± 5 degrees F for 24 hours.

b. Condition for 24 hours at 77 degrees F ± 5 degrees F and 40 to 60 percent relative humidity.

c. Test to insure that the equipment meets specified performance of 4.7.

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d. Place the equipment in its normal operating position with all covers installed in a test chamber maintained at 77°F ± 5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65°F ± 5°F and shall fall freely at a rate of two inches per hour. Maintain this condition for one hour.

e. With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour, against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for one hour.

f. Rotate the equipment 90 degrees horizontally to expose a second side of the component to the wind and rain. Maintain this condition for one hour.

g. Continue as in 4.14f until all four sides of the equipment have been exposed to the wind and rain, each side for one hour.

h. Immediately after completion of the foregoing, repeat the test specified in 4.7 without removal from the chamber but with the rain and wind discontinued. Complete the test within twenty minutes. The equipment shall meet specified performance (see 3.11.4).

i. Open the equipment. There shall be no visible evidence of leakage.

4.15 Vibration (Internal). The equipment shall be subjected to the following test:

a. The equipment with covers removed shall be secured to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that the vibration within the test item can be observed and measured.

b. The equipment shall be vibrated successively in three mutually perpendicular directions over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall not be less than 0.030 inch. In each of the three directions, the frequency shall be changed in steps of 1 Hz and maintained for at least 10 seconds. Vibration amplitudes shall be measured by optical or other means, provided the vibration of the test item is not affected by the measurement.

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c. The equipment shall have no resonance in the frequency range of 10 to 55 Hz that exceeds twice the amplitude of the applied vibration.

d. Electrical test data for the tests specified in 3.11.5.1 shall be taken before and after this test.

4.16 Shock, Bounce, Vehicular. The equipment shall be subjected to Test Method 514.2, Procedure IX, Part 2 of MIL-STD-810. Pretest data is required and is specified in 3.11.5.2. During this test, the equipment shall be secured in a sturdy metal simulated rack meeting the dimensional requirements of a standard 19-inch rack as specified in MIL-STD-189. The equipment shall be positioned in its normal operating position and supported against cantilever action of the case. The simulated rack containing the equipment shall be mounted on a vehicular adaptor place on the package tester. At the conclusion of the test, the equipment shall meet the requirements of 3.11.5.2 and shall contain no damage.

4.17 Shock (Bench-Handling). The chassis-and-front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing.

a. Tilt up the assembly through an angle of 30 degrees using one edge of the assembly as an axis and permit the assembly to drop back freely to the horizontal. Repeat using other practicable edges of the same horizontal face as axis for a total of four drops.

b. Repeat a. with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing. (see 3.11.5.3)

c. Equipment shall conform to specified performance of 3.11.5.3 after the test is completed.

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4.18 Fungus. The equipment shall be subjected to the fungus test specified in Method 5081, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (6.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.11.7 shall constitute failure of this test.

4.19 Salt fog. The equipment shall be subjected to the salt fog test specified in Method 5091, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.11.6 shall constitute failure of this test.

4.20 Inspection for Interchangeability (see 3.6). The dimensions and electrical performance listed below shall be gaged or measured to determine conformance to the interchangeability requirement of 3.6. When a listed dimension is not within specified limits, it shall be considered a major defect.

4.20.1 Mechanical (dimensions). Gage points covered by gage list SC-GL-3232L4 shall be measured.

4.20.2 Electrical Performance (sub-assemblies). The following subassemblies shall be tested to determine compliance with the performance requirements specified in the applicable test requirement specification (TRS) drawings referenced in 4.10. Subassemblies shall be selected from end equipments that have been subjected to and met the Group A inspection.

18A1A	Power Supply converter
18A2	Panel assembly, generator, 20 and 1600 cycles
18A3B	Panel assembly, Channel Unit, with Plug Supervision

4.21 Visual and Mechanical Inspection. The equipment shall be examined for the defects listed in MIL-STD-252 for conformance with MIL-STD-454 and the requirements of 3.3 and 3.15.

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4.22 Ancillary Items. Ancillary items required for testing by this specification shall be furnished by the contractor in the quantities required to accomplish the testing herein. The ancillary items shall be as follows:

4.22.1 Interconnecting Cables. The interconnecting cables used with the equipment are as follows:

4.22.1.1 Cable Assembly, Power CX-11173( )/U. The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531015.

4.22.1.2 Cable Assembly, Special Purpose CX-9088( )/U. The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531022.

4.22.2 Mounting Bracket Assembly, Including the following:

2 each mounting bracket per SM-B-528626
8 each screws per MS-35276-263
8 each washers per SM-B-528696
8 each washer, lock per MS-35338-81

4.22.3 Panel Assembly, Dummy Test. The dummy test panel assembly shall be constructed in accordance with and meet the requirements of SC-DL-528640.

4.22.4 Outer Front Cover. The cover shall be constructed in accordance with, and meet the requirements of, SM-C-528566. This cover shall be used in the rain test.

4.23 Quality Conformance Inspection of Preparation for Delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be at Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Converter, Telephone Signal, CV-1548( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Each Converter, Telephone Signal CV-1548( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application. None required.

5.1.1.4 Unit Packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. Each technical literature manual shall be packaged method IC-1.

5.1.1.4.2 Converter, Telephone Signal CV-1548( )/G. Each converter, telephone signal shall be individually packaged method III as follows: Place the spare fuses and spare lamps within the clips provided for on the front panel. Cushion each unit on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. Converter, Telephone Signal CV-1548( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be Level A, B, or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2b).

5.2.1 Level A.

5.2.1.1 Palletized Load. A quantity of Converter, Telephone Signal CV-1548( )/G, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked

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load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.1.2 Less than Palletized Load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B.

5.2.2.1 Palletized Load. A quantity of Converter, Telephone Signal CV-1548( )/G, packaged as specified in 5.1 shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than Palletized Load. When quantities per destination are less than a pallet load, a quantity of Converter, Telephone Signal CV-1548( )/G, packaged as specified in 5.1 shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

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5.2.3 Level C.

5.2.3.1 Palletized Load. A quantity of Converter, Telephone Signal CV-1548( )/G, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than Palletized Load. When quantities per destination are less than a pallet load, a quantity of Converter, Telephone Signal CV-1548( )/G packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. Converter, Telephone Signal CV-1548( )/G is intended for use with carrier terminal equipments which do not have internal signalling or hybrid facilities. Signalling equipment (20 Hz) in the converters is compatible with the signalling circuits of Manual Telephone Switchboards SB-86( )/P, Central Office, Telephone Manual AN/TTC-7( ), and Telephone Sets TA-312( )/PT.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation and packaging and packing. (See Section 5).

c. When rough handling and functional test are required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required.

g. First Article Inspection.

h. Quantity of tools and running spare parts required.

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6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.3.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C Inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action. (See 4.5.3.3).

6.4.1 Initial Shipments. Initial shipments of end items will be contingent only upon successful completion of all Group C tests. (See 4.5.3)

6.5 Location of Operational Inspection. It is desirable that the operational system test (4.7) be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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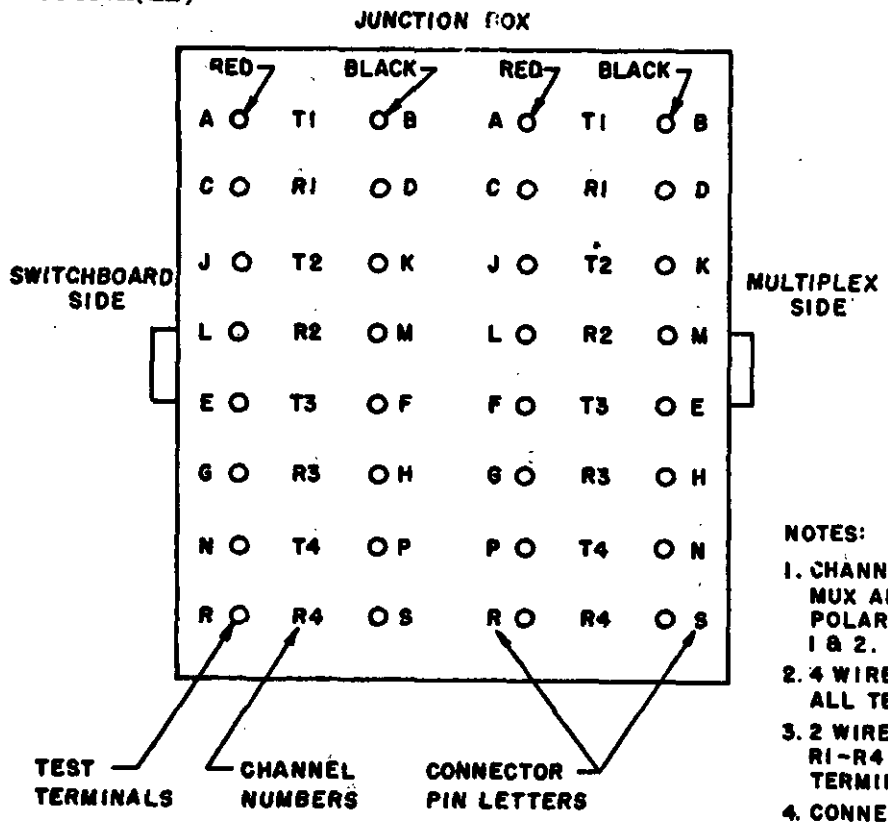
6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CV-1548W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
Army-EL

Preparing Activity
Army-EL
Project Number 5805-A226

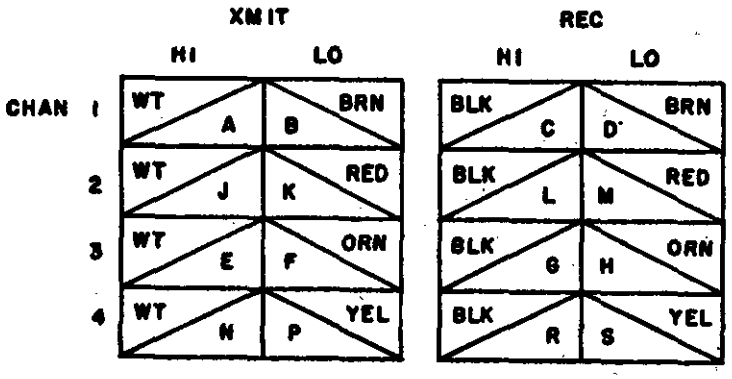
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NOTES:

1. CHANNEL 3 & 4 TRANSMIT-MUX ARE OPPOSITE POLARITY FROM CHANNEL 1 & 2.
2. 4 WIRE CONNECTION — USE ALL TERMINAL PAIRS.
3. 2 WIRE CONNECTION — USE R1-R4 OF SWITCHBOARD TERMINAL PAIRS.
4. CONNECT RED TO "TIP" FOR SB-86/P; BLACK TO "RING"
5. CONNECT RED TO "RING" FOR AN/TTC-7 AND OTHER COMMON BATTERY SWITCH-BOARDS; BLACK TO "TIP"
6. ALL RESISTORS SHOWN ARE 600 ± 6 OHMS UNLESS DENOTED OTHERWISE.
7. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
8. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.

CX — 9088( )/U
CABLE ASSEMBLY

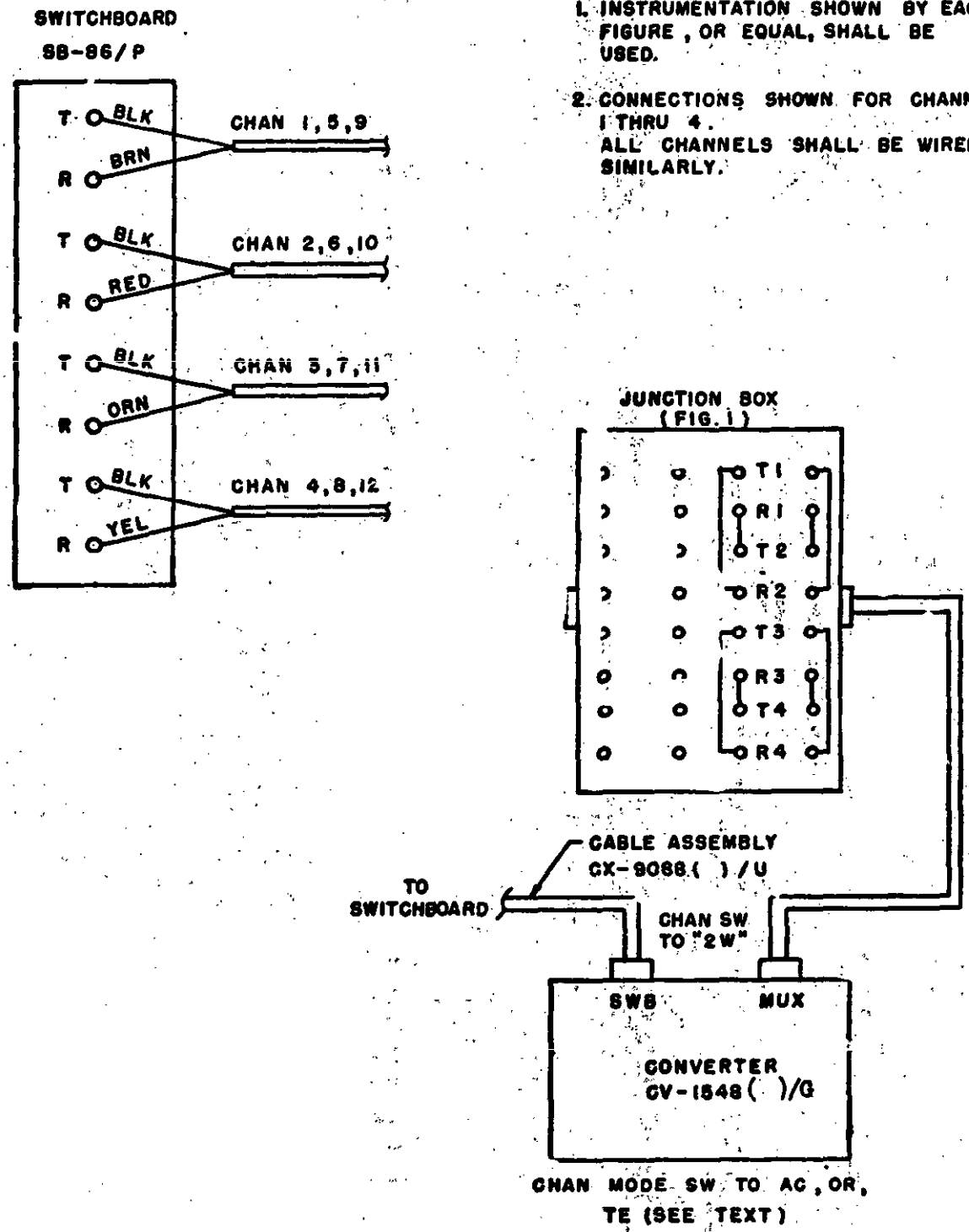


WIRING DIAGRAM
FIG. 1

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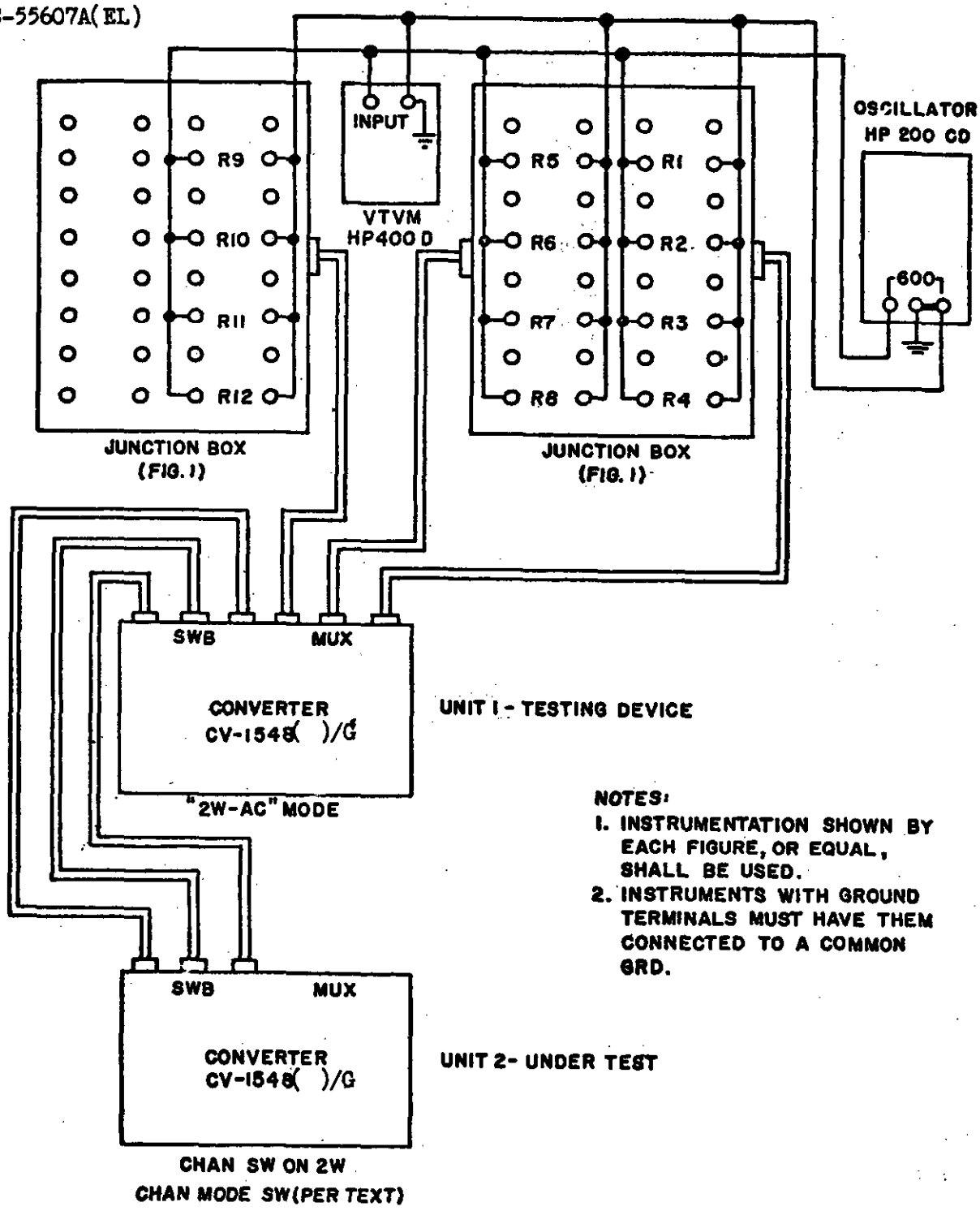
NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. CONNECTIONS SHOWN FOR CHANNELS 1 THRU 4.
   ALL CHANNELS SHALL BE WIRED SIMILARLY.



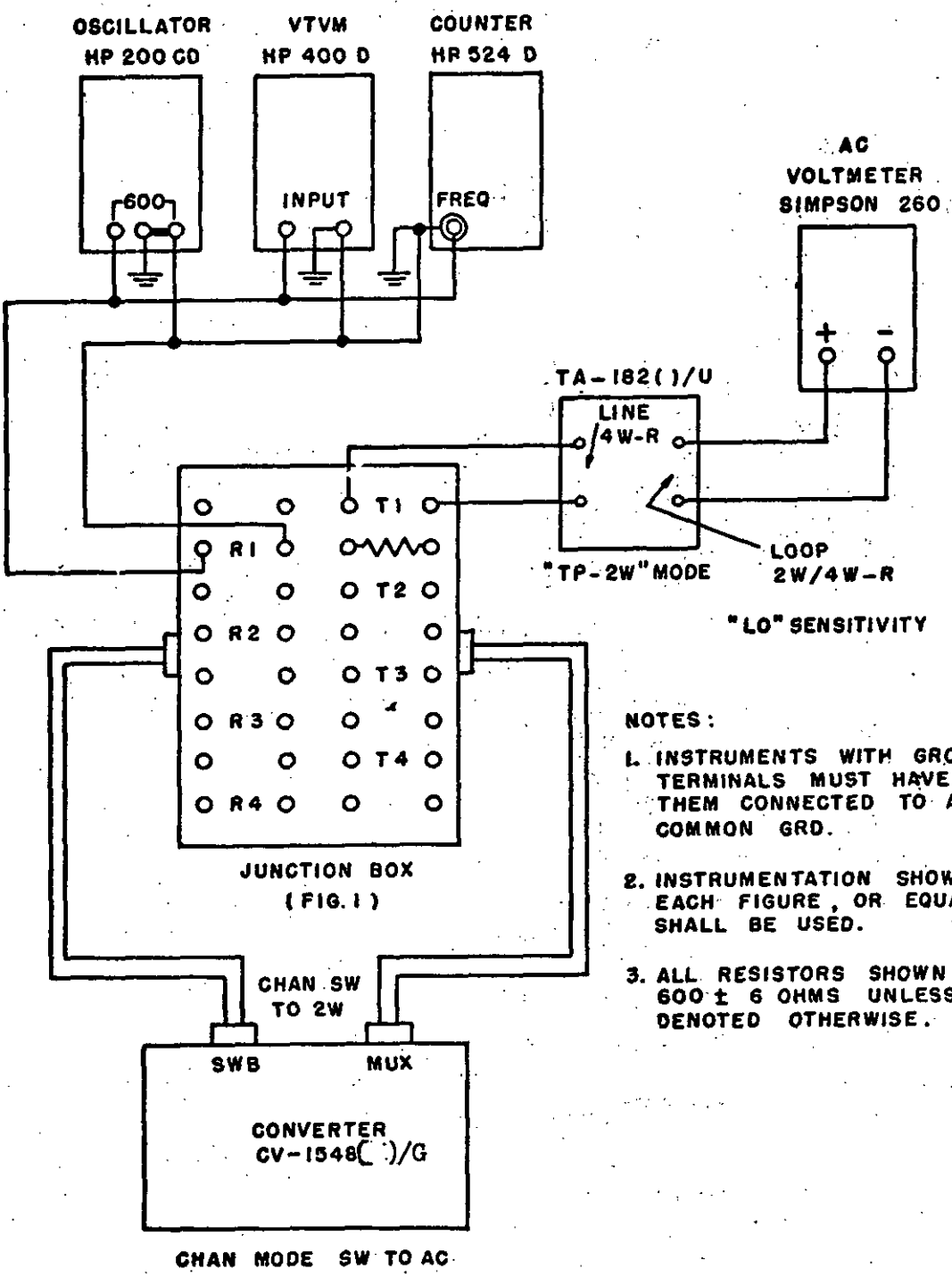
SWITCHBOARDS INTERFACE TEST
FIG. 2

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2 W CIRCUIT PROTECTION TEST
FIG. 3

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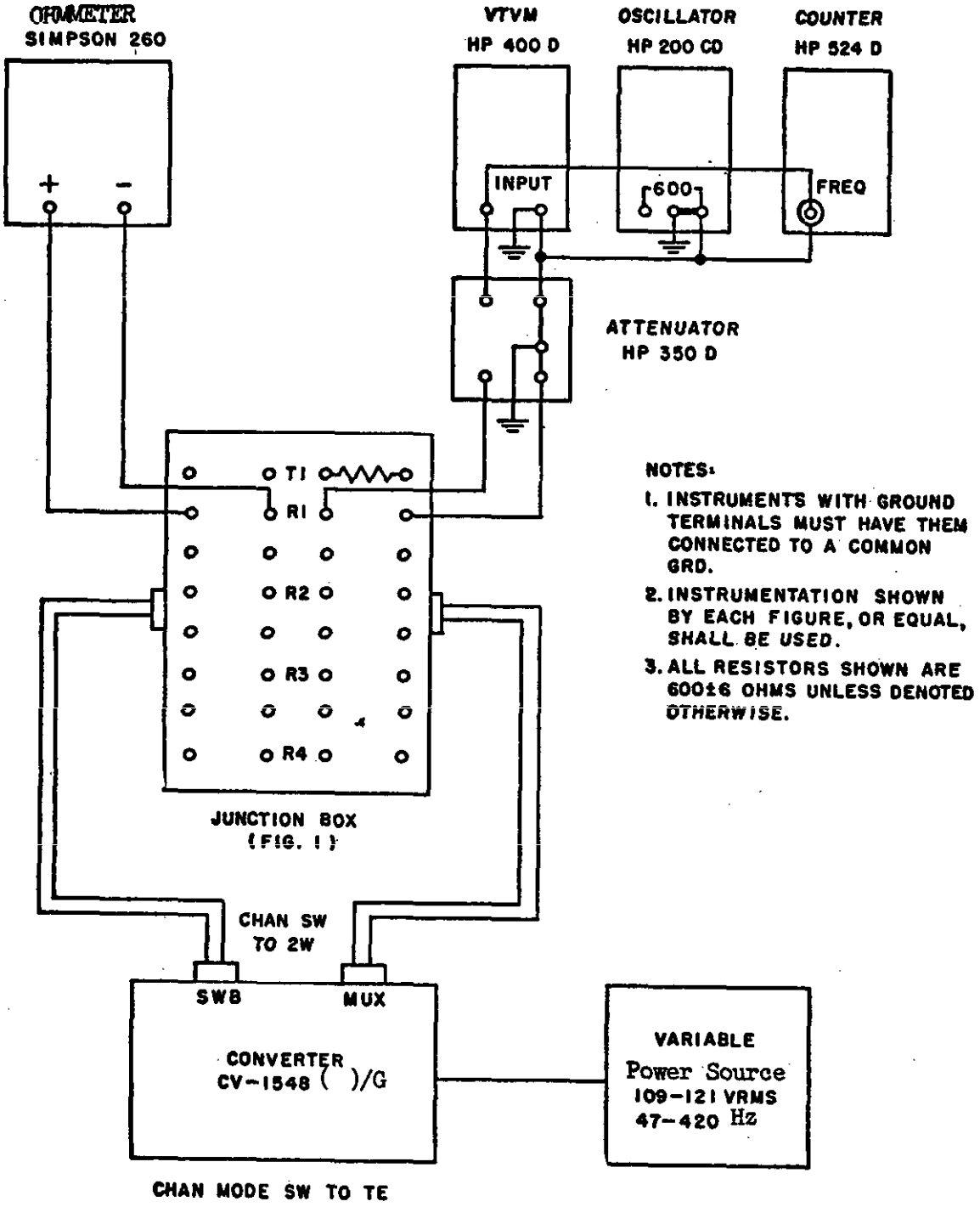


NOTES:

1. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
2. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
3. ALL RESISTORS SHOWN ARE 600 ± 6 OHMS UNLESS DENOTED OTHERWISE.

20 Hz SENSITIVITY TEST
FIG. 4

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NOTES:

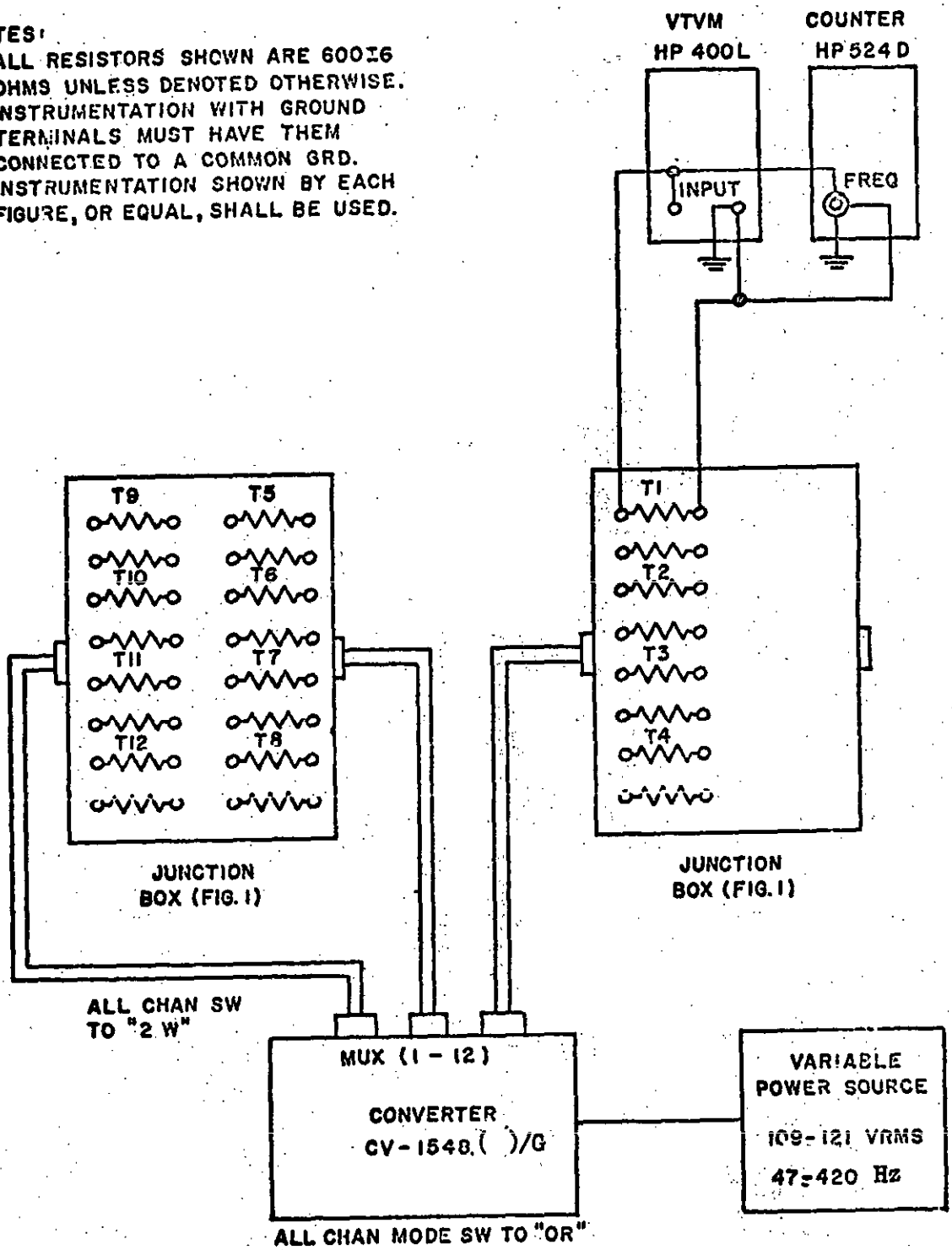
1. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
2. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
3. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.

1600 ‹ Hz SENSITIVITY TEST

FIG. 5

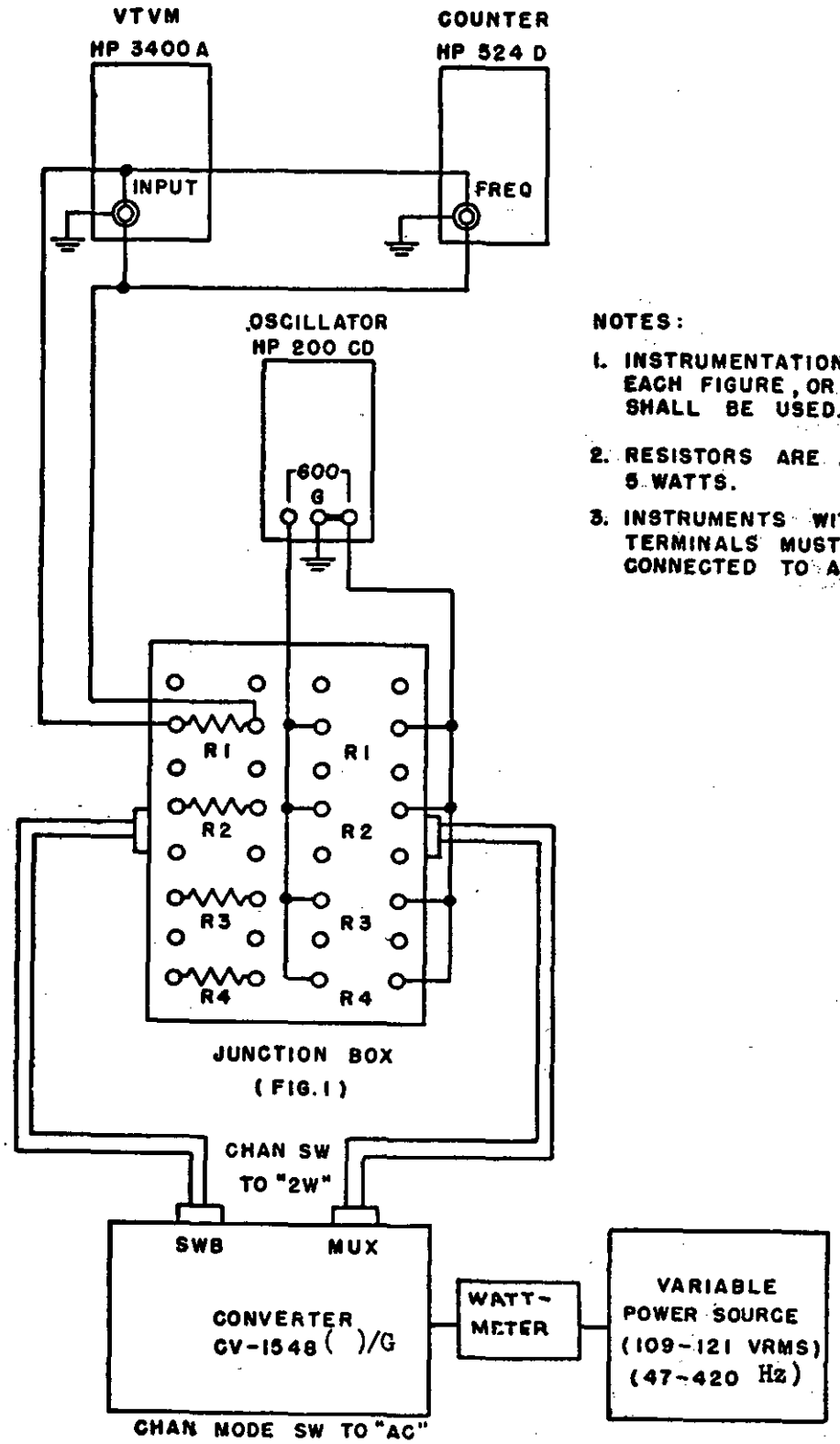
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NOTES:
1. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.
2. INSTRUMENTATION WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.



1600 Hz GENERATOR TEST
FIG. 6

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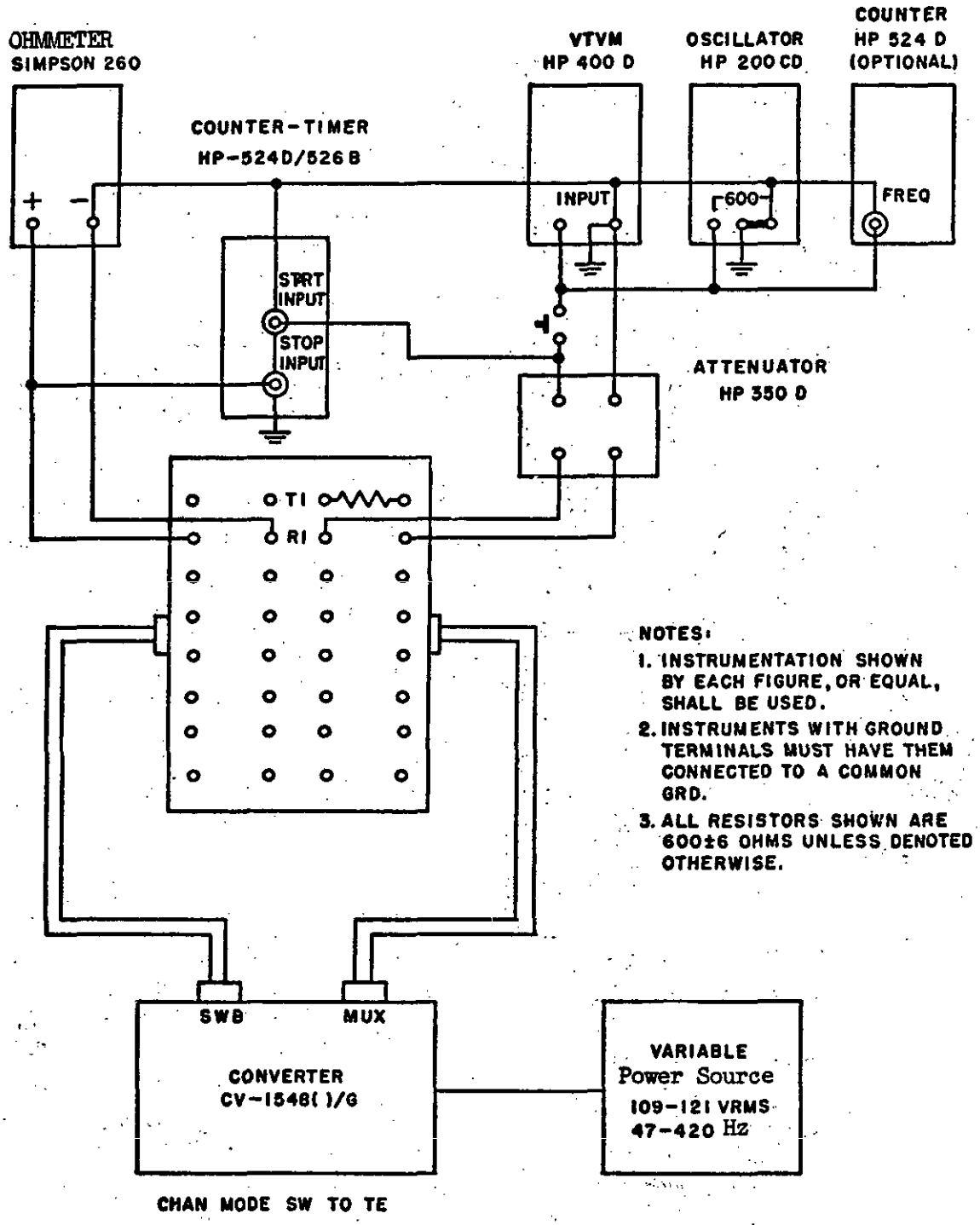
NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. RESISTORS ARE 2500 OHM, 5 WATTS.
3. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.

20 Hz GENERATOR TEST

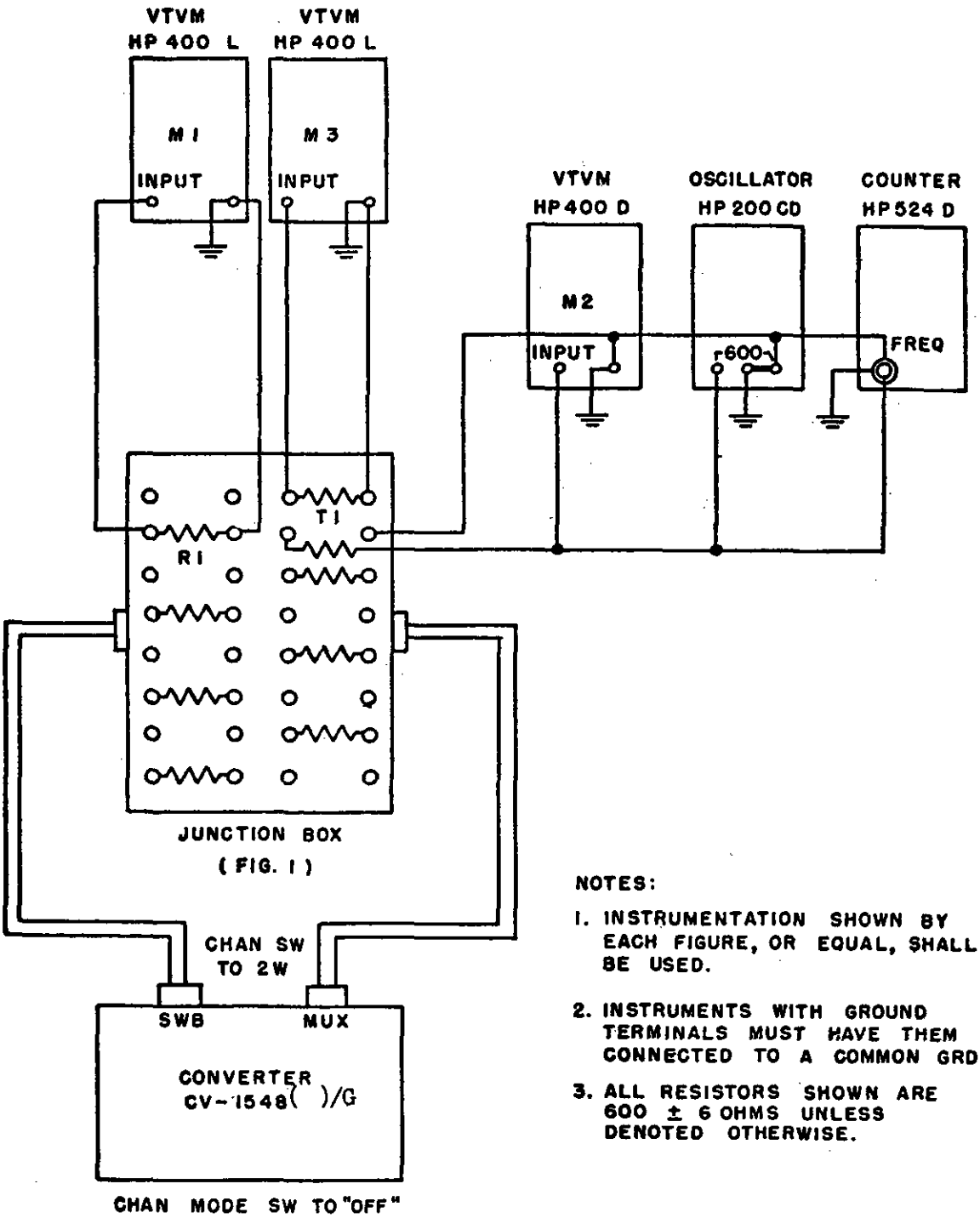
FIG. 7

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GUARD CIRCUIT TEST
FIG. 8

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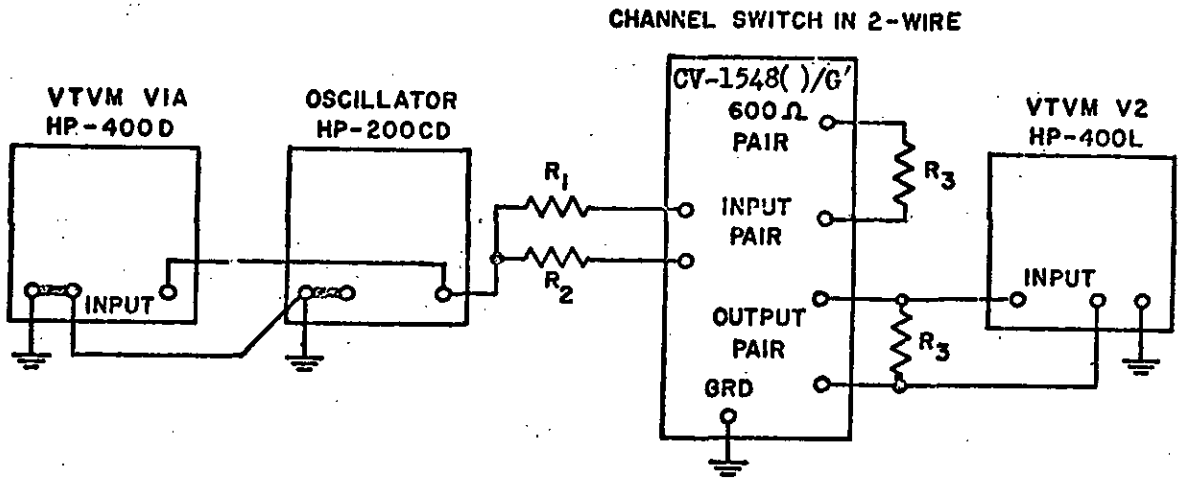


NOTES:

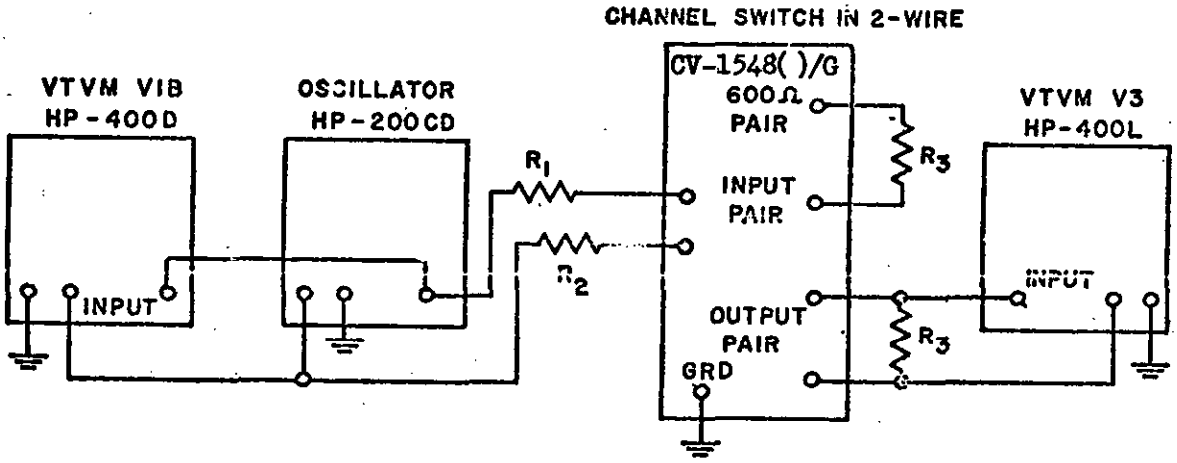
1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 600 ± 6 OHMS UNLESS DENOTED OTHERWISE.

HYBRID TESTS
FIG. 9

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OUTPUT WITH UNBALANCED INPUT
FIG. 10A

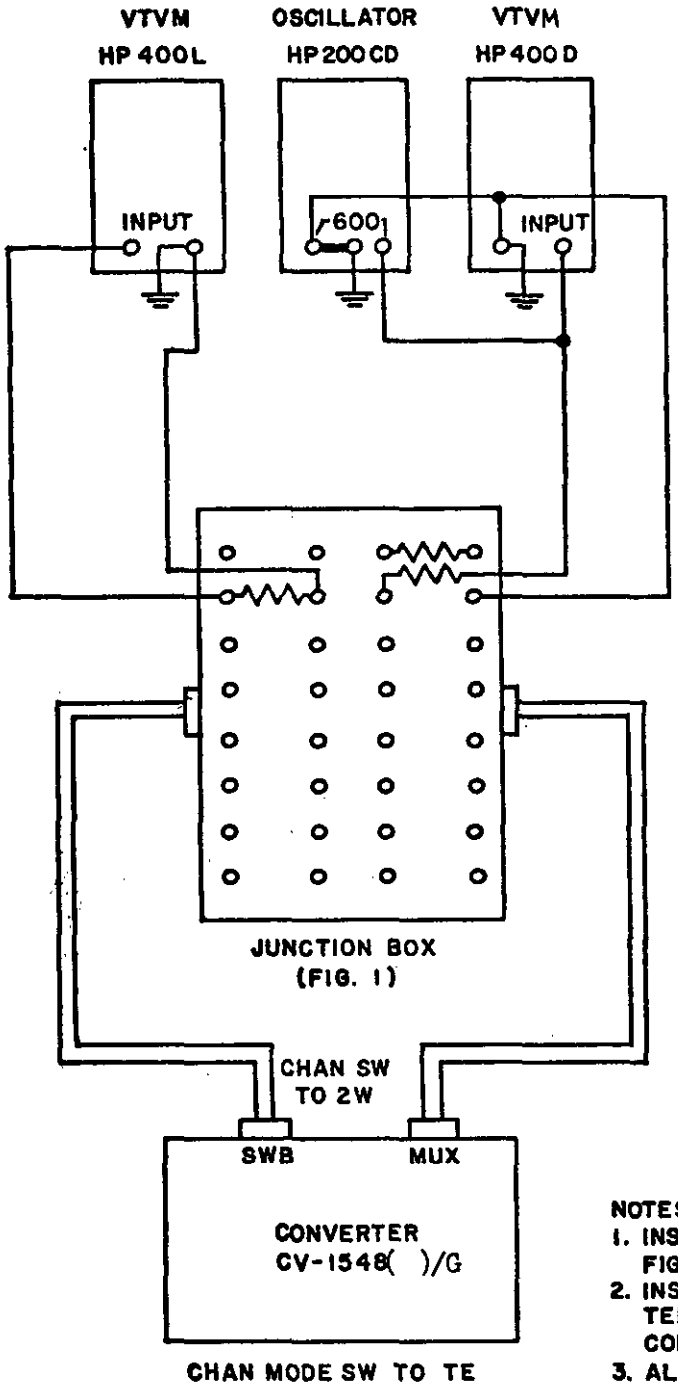


OUTPUT WITH BALANCED INPUT
FIG. 10B

NOTES:
1. R1 AND R2 ARE 300 OHMS RESISTORS MATCHED WITHIN 0.3 OHMS OF EACH OTHER.
2. R3 IS 600Ω.
3. ALL RESISTORS ARE ±1%.
4. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
5. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.

LONGITUDINAL BALANCE TEST
FIG. 10

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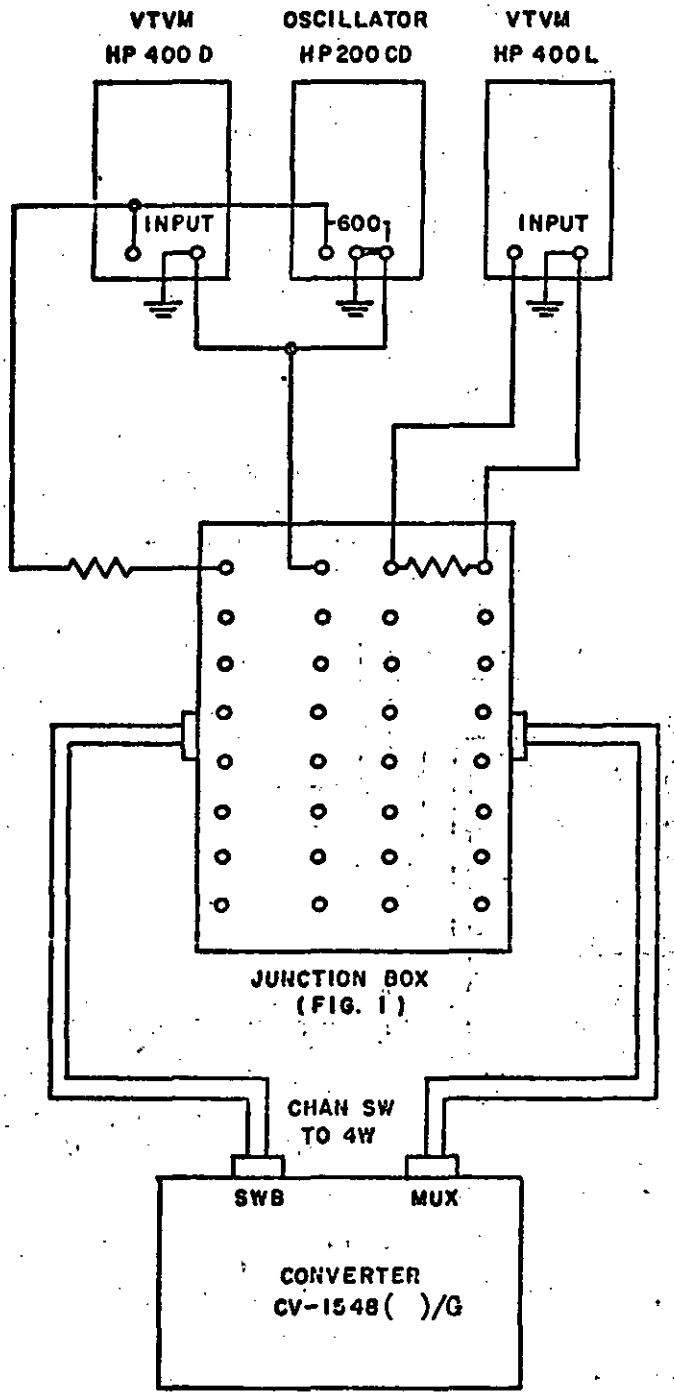


NOTES:
1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.

CHANNEL FREQUENCY RESPONSE (4W-2W)

FIG. 11

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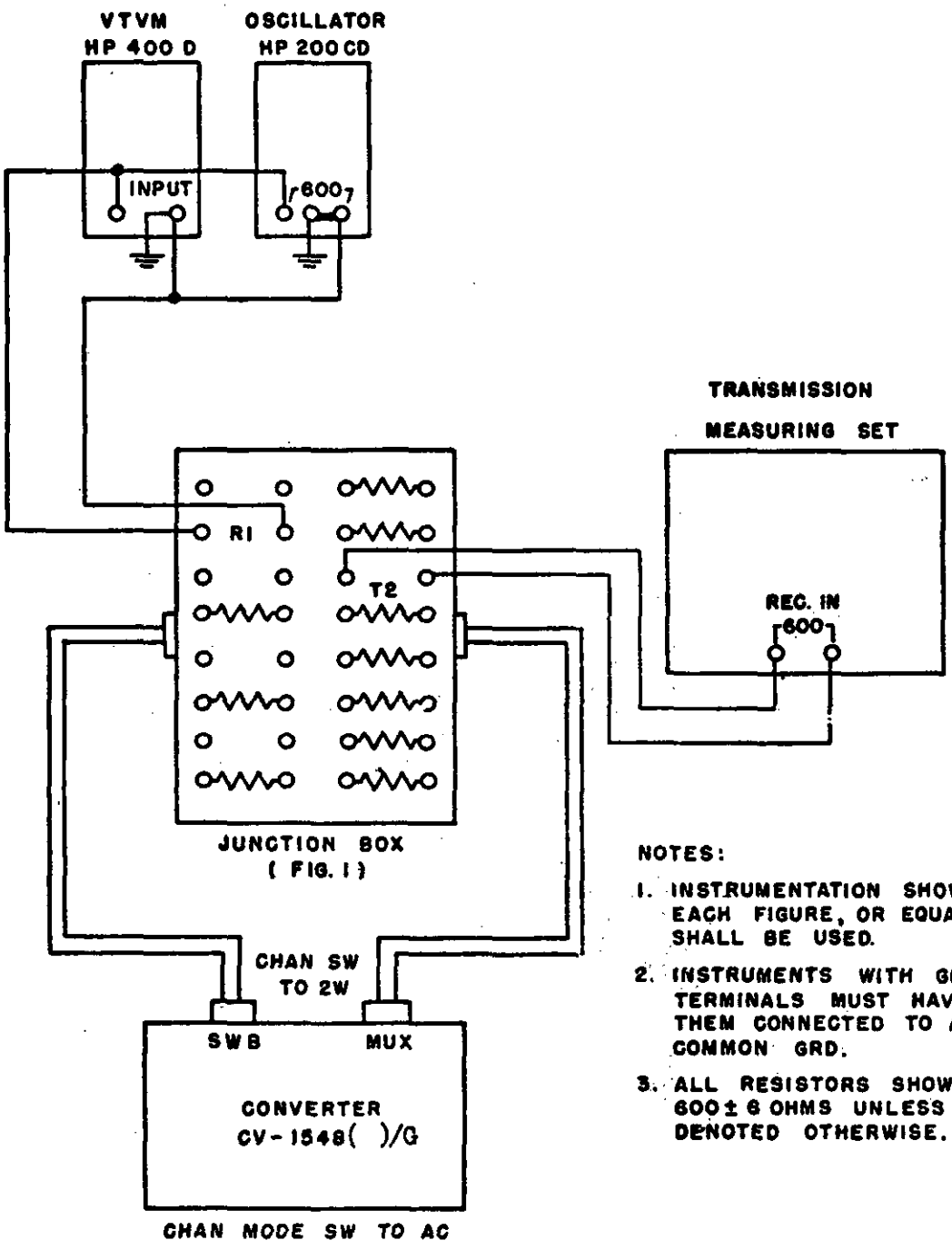
NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 300±6 OHMS UNLESS DENOTED OTHERWISE.

CHANNEL FREQUENCY RESPONSE (4W–4W)

FIG. 12

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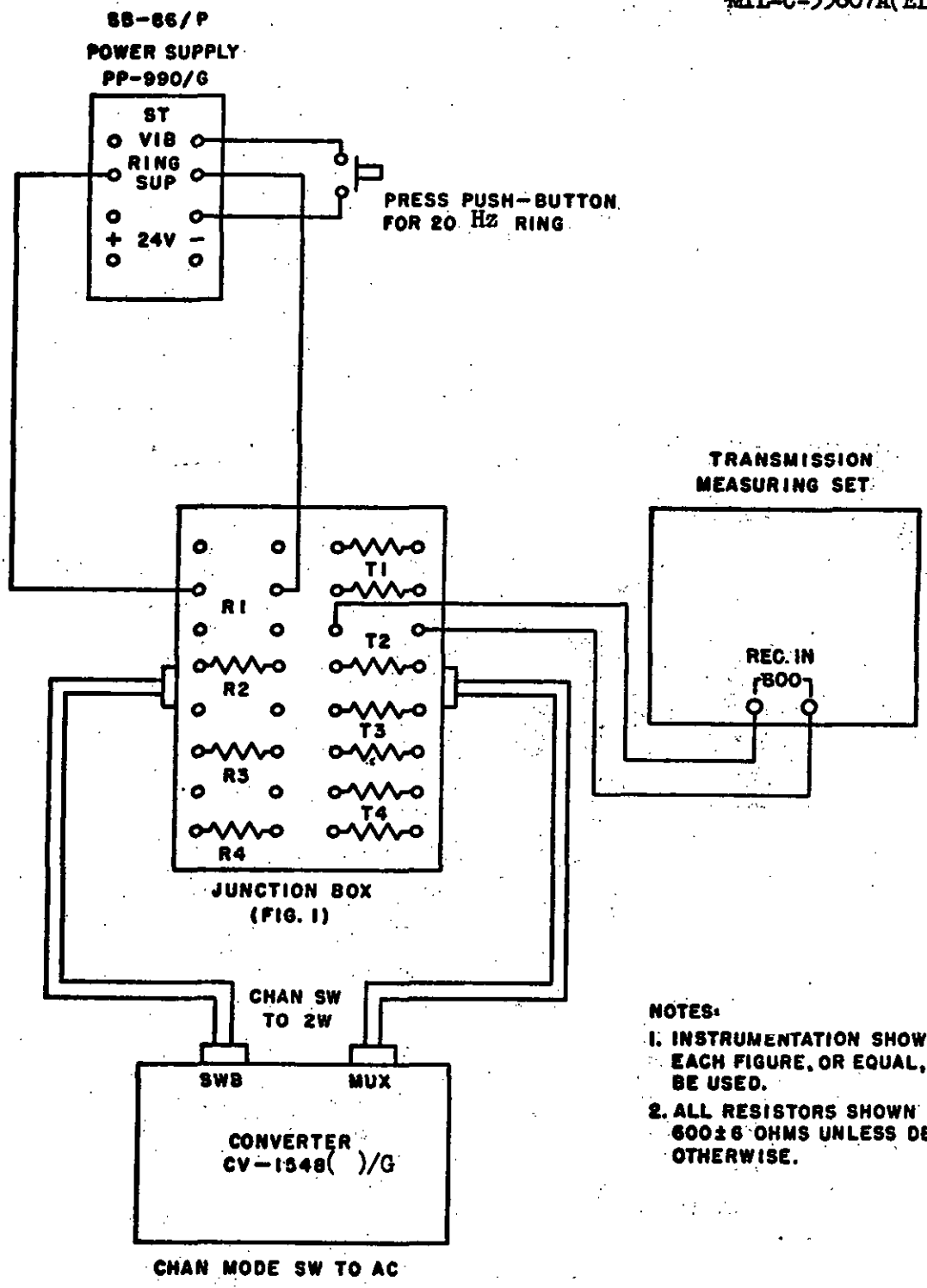


NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 600 ± 6 OHMS UNLESS DENOTED OTHERWISE.

1000 Hz CROSSTALK
FIG. 13

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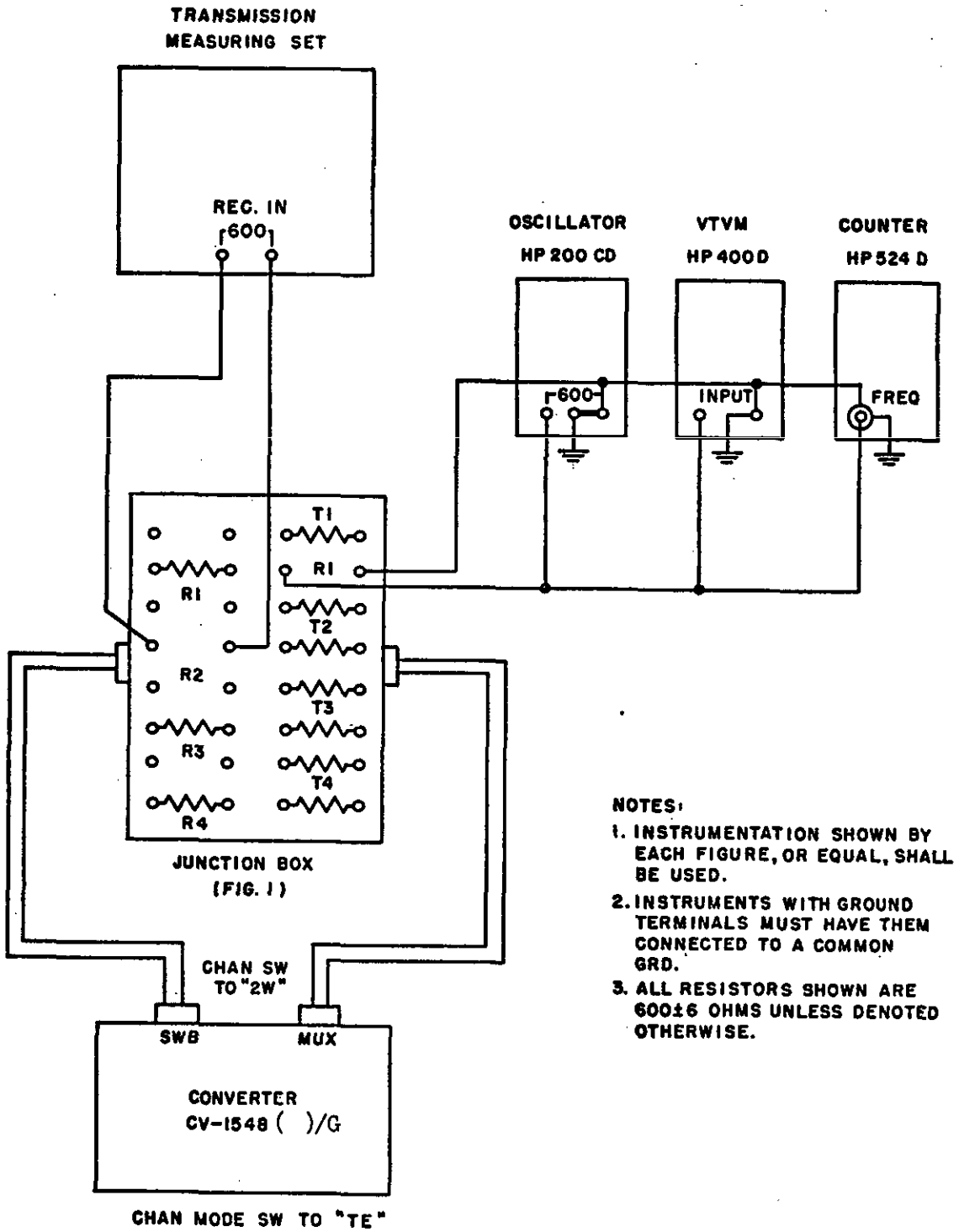


NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.

20 Hz CROSSTALK
FIG. 14

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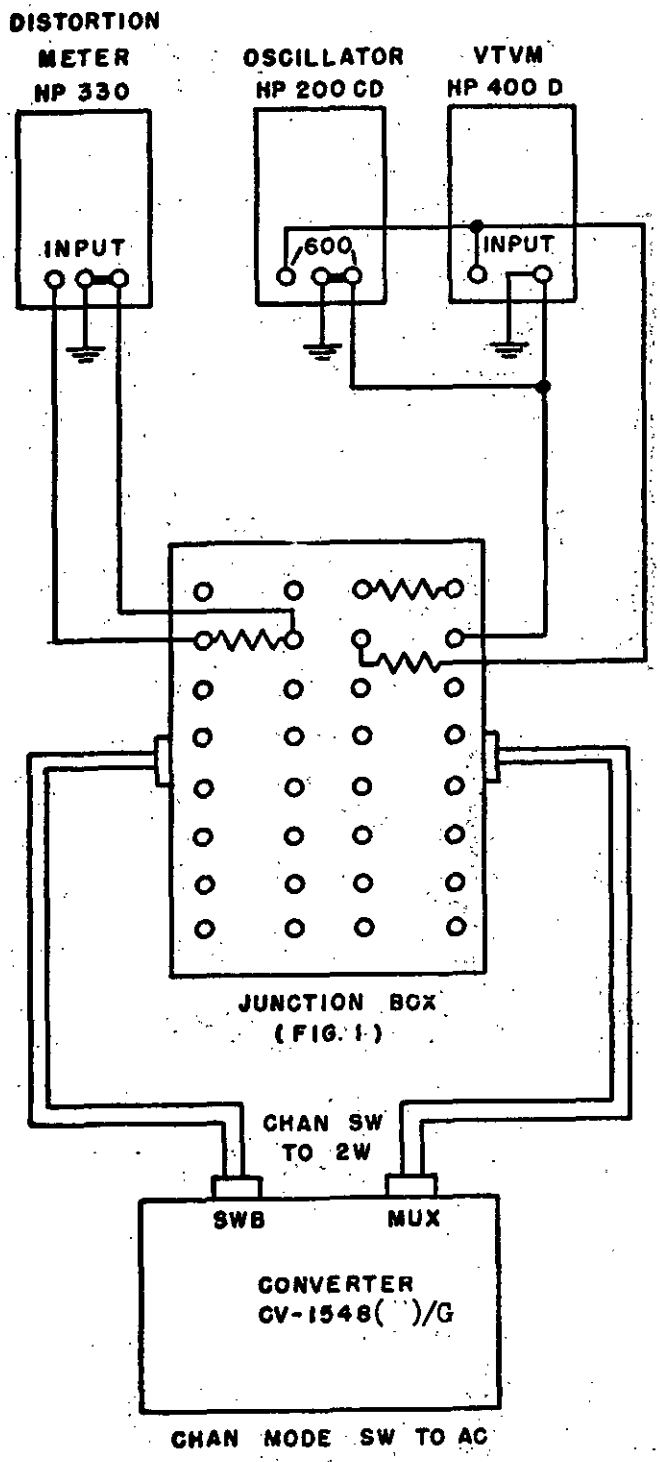
NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.

1600 Hz CROSSTALK

FIG. 15

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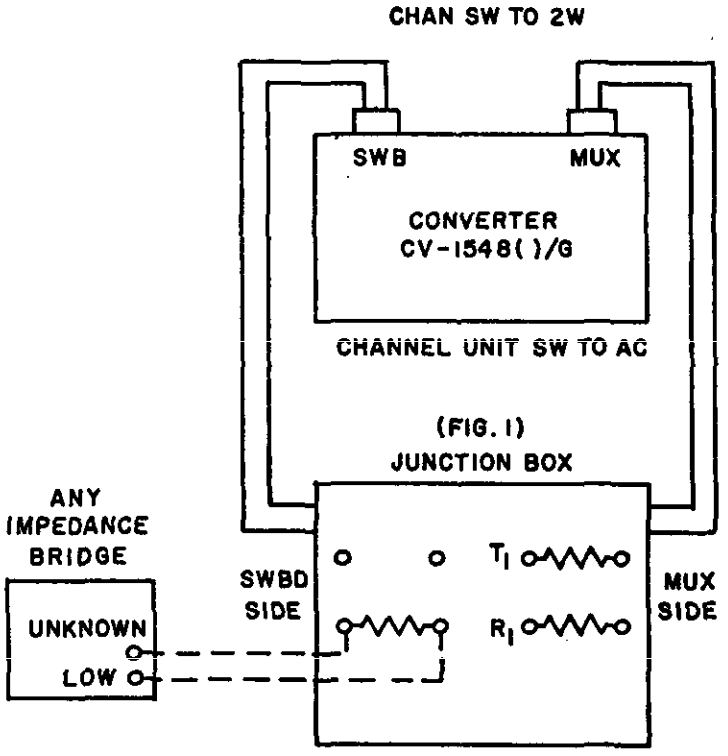


NOTES:

1. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
2. INSTRUMENTS WITH GROUND TERMINALS MUST HAVE THEM CONNECTED TO A COMMON GRD.
3. ALL RESISTORS SHOWN ARE 600 ± 6 OHMS UNLESS DENOTED OTHERWISE.

SIGNAL DISTORTION (4W-2W)
FIG. 16

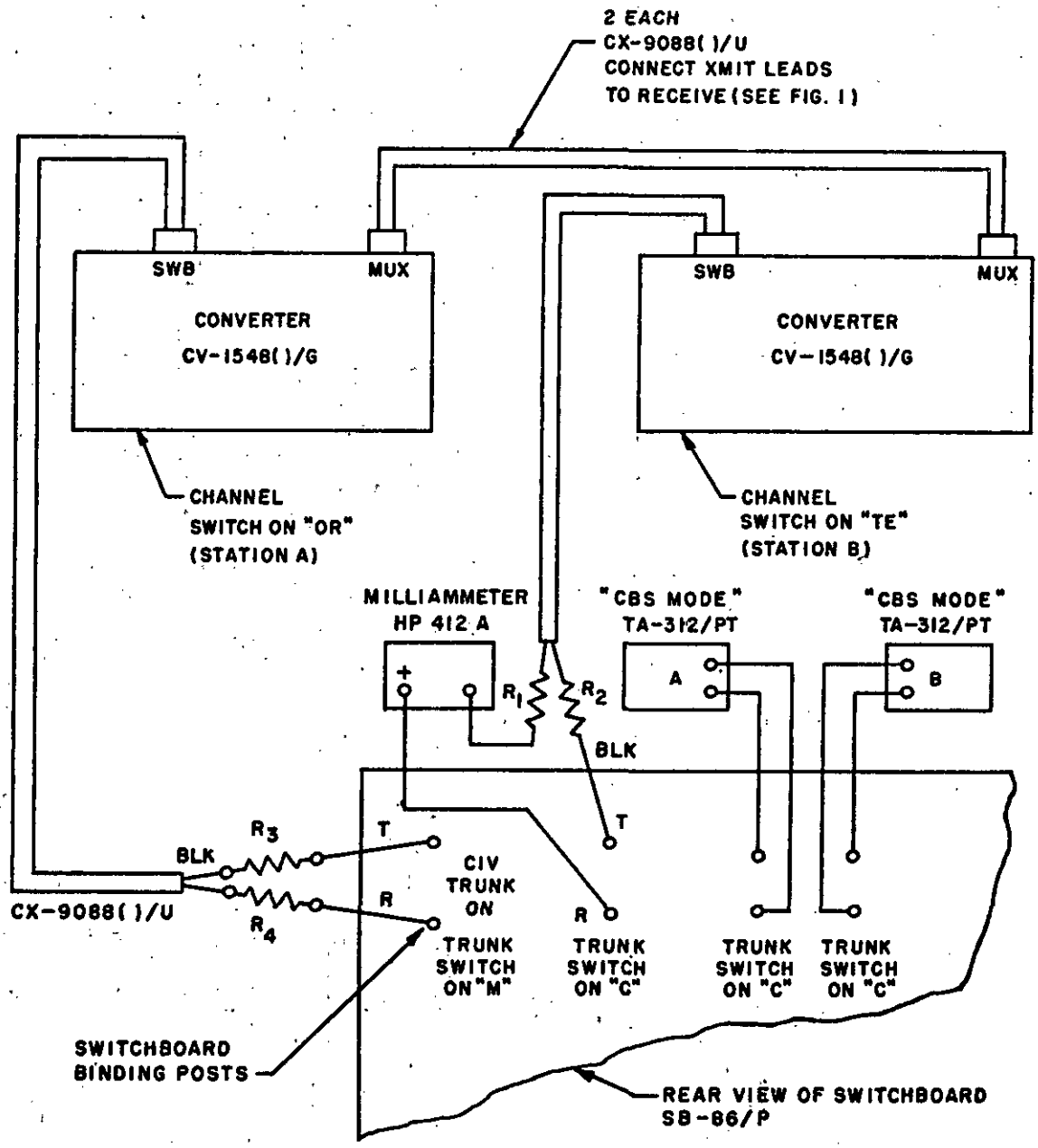
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NOTES:
1. CONNECTIONS SHOWN FOR CHANNEL 1. ALL CHANNELS ARE WIRED SIMILARLY.
2. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.
3. ALL RESISTORS SHOWN ARE 600±6 OHMS UNLESS DENOTED OTHERWISE.

IMPEDANCE TEST
FIG. 17

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NOTES:

1. NO CONNECTIONS ON TRANSMIT LEADS.
2. R_1, R_2, R_3 & R_4 RESISTORS (R) ARE 150 ± 15Ω, 1 WATT.
3. INSTRUMENTATION SHOWN BY EACH FIGURE, OR EQUAL, SHALL BE USED.

PLUG SUPERVISION TEST

FIG. 10

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55607A(EL) CONVERTER, TELEPHONE SIGNAL CV-1548( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. *GIVE PARAGRAPH NUMBER AND WORDING.*

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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